

# Sample-Size Dependent Representation Drift in Moirai

## Fine-Tuning:

### A Case Study on ETT Forecasting

Anonymous Author(s)

May 4, 2026

#### Abstract

We present a case study of Moirai fine-tuning [26] on ETT forecasting benchmarks, documenting a **cost–benefit gap**: fine-tuning imposes geometric cost (CKA drops from 0.949 at  $n=500$  to  $\approx 0.52$  at  $n=10,000$ ) while producing a task-aligned functional benefit (Ridge  $\Delta R^2 > 0$  on the trained 96-step head in 10/10 seeds on ETTh2 and ETTm2;  $\Delta R^2 \leq 0$  on untrained heads). Task forgetting is a protocol-dependent consequence:  $-5.3 \pm 6.2\%$  under early-stopped checkpointing (7/10 negative),  $+7.5 \pm 8.2\%$  under final-epoch (dominated by late-epoch overshoot in 3 seeds). We distinguish three sample-size regimes as a descriptive taxonomy: low- $n$  forgetting, high- $n$  improvement (ETTh2/ETTM2), and a boundary case (ETTh1: task resolves but  $\Delta R^2$  is 2/10 positive). Three controls rule out confounds: a frozen-encoder control ( $\Delta R^2 = -0.048 \pm 0.008$ ,  $14 \times$  smaller than fine-tuned) rules out data-distribution effects; a random-init control (5 seeds,  $\Delta R^2 = -0.199 \pm 0.205$ , 4/5 negative) rules out “any encoder fine-tuned on this objective”; a layer-wise unfreeze (top 3 of 6 layers sufficient for  $\Delta R^2 > 0$ , 3/3 seeds) localises the effect.

**Contribution tiering.** We distinguish methodological contributions (backbone-agnostic: a value-gate framework for pre-training quality, LoRA-Large  $10 \times$  LR rescue recipe cross-dataset validated on ETTh1/ETTh2/ETTM2, controlled-comparison designs, and a uni2ts PackedStdScaler patch) from empirical findings (Moirai-ETT specific: the three-regime taxonomy, per-seed  $\Delta R^2 > 0$  invariant on ETTh2/ETTM2). Five non-Moirai backbones (Chronos-T5-Small/Base/Large, TimesFM-2.5, MOMENT-base) tested across nine ETT  $\times$  backbone cells all gate-fail (ZS  $< 20\%$  over Linear), a finding about ETT coverage in their pre-training corpora independent of the Moirai-specific results. **Practitioner takeaway:** Use LoRA as default for Moirai adaptation; on Moirai-Large, reduce LR by  $10 \times$  before escalating rank.

## 1 Introduction

Moirai [26] is a foundation time-series model trained on billions of time points that achieves strong zero-shot performance on forecasting benchmarks. A natural next step is to fine-tune it on domain-specific tasks: anomaly detection, classification, or short-horizon forecasting with limited labelled data. We present a Moirai case study on ETT forecasting benchmarks that characterises what happens to pre-trained representations during fine-tuning.

**Representation drift during fine-tuning.** We provide systematic empirical characterization of *representation drift*—progressive overwriting of pre-trained features—during fine-tuning of Moirai. The drift–utility dissociation is established on the Moirai family (Small 13.8M, Base 91M, Large 311M). Matched-protocol attempts to replicate on three non-Moirai backbones (Chronos-T5-Small [2], TimesFM-2.5-200M [4], MOMENT-1-base [8]) gate-fail on all nine ETT  $\times$  backbone cells tested (7 original + 2 new Chronos capacity cells); we report this as a substantive finding about backbone-specific ETT pre-training coverage (§5, Appendix A). Catastrophic forgetting during fine-tuning is well-studied in continual learning [12, 16] and NLP [19]; our contribution is a systematic case-study diagnosis for Moirai, with controlled experiments isolating drift from overfitting and a spectrum of mitigations. We use “catastrophic forgetting” to mean degraded performance on distributions the model originally handled well—consistent with continual learning usage where the “previous task” is the pre-training objective.

**Distinguishing drift from overfitting.** Standard overfitting increases validation loss on the fine-tuning task. Representation drift instead degrades zero-shot performance on tasks the model previously handled well, while training loss continues to decrease monotonically. Our frozen-encoder control (same data, same epochs, CKA=1.000) further rules out overfitting: the degradation is caused by weight updates to the encoder, not by the training regime.

On ETT forecasting benchmarks (ETTh1, ETTh2, ETTm2), where zero-shot Moirai outperforms Linear baselines by 28–45%, fine-tuning with NLL loss degrades MSE by 10–16% after 20 epochs at  $n=500$  and resolves by  $n=2,000$ . On IoT (sub-random ZS AUC), CKA drops to 0.31–0.45 but task performance improves from a near-null baseline: the drift is irrelevant because there were no valuable features to forget.

**Why this matters.** Unlike language models, where drift manifests as degraded generation quality, time-series drift directly harms the distributional predictions that underpin anomaly scoring and probabilistic forecasting. Our study isolates drift’s task-level consequences on Moirai and their dependence on the value of pre-trained features: harmful on ETT in the low-data regime, benign once  $n$  is large enough for Moirai to learn replacements, and irrelevant on IoT where ZS features have no value to begin with.

**Experimental approach.** We design a controlled diagnosis with fine-tuning conditions (B: NLL-only, C: NLL+SupCon) and standard mitigations (D: frozen encoder, E: LoRA [10], F: L2-SP [15], G: EWC [12]) across three Moirai model sizes (Small 13.8M, Base 91M, Large 311M). We deploy this on two settings: (1) ETT forecasting benchmarks [29] (ETTh1, ETTh2, ETTm2), chosen because zero-shot Moirai outperforms Linear baselines by 28–45%, ensuring pre-trained features are demonstrably useful; and (2) IoT anomaly detection as a *negative control* (Moirai ZS AUC sub-random; we do not claim IoT as a contribution, only as validation that the value-gate correctly suppresses the dissociation test when ZS features have no value). Both settings use CKA, linear probing, and weight drift as representation-change diagnostics. We report forgetting under two protocols (CUDA 10-seed deterministic early-stopping and final-epoch  $k=5$ ); the sign of the  $n=10k$  headline differs between them (+7.5% vs. −5.3%), so we lead on the more robust  $\Delta R^2 > 0$  majority and treat forgetting sign as a secondary, protocol-dependent outcome.

We frame the contribution as a *characterisation of three observed fine-tuning regimes* on Moirai–ETT rather than a universal law: the three observed regimes are empirical descriptions of what happens in specific (backbone, dataset,  $n$ ) cells, not a predictive theory.

**Contributions.** We distinguish three tiers:

1. **Methodological contributions (backbone-agnostic).** These contributions generalize beyond Moirai–ETT:
  - (a) **Value-gate framework.** A criterion (ZS advantage  $>20\%$  vs. Linear) for screening whether pre-trained features are worth preserving. IoT (sub-random ZS) validates the gate as a negative control.
  - (b) **LoRA-Large  $10\times$  LR rescue recipe.** Cross-dataset validated on ETTh1/ETTh2/ETTM2 (3-5 seeds each). Rank escalation does not rescue; LR reduction does.
  - (c) **Controlled-comparison design.** Frozen-encoder control ( $14\times$  smaller  $\Delta R^2$  than fine-tuned), random-init control (5 seeds,  $\Delta R^2 = -0.199 \pm 0.205$  vs. pre-trained  $+0.668 \pm 0.226$ ), and layer-wise unfreeze (top 3 of 6 layers sufficient for  $\Delta R^2 > 0$ ) isolate distinct confounds.
  - (d) **uni2ts PackedStdScaler patch.** Fixes silent autograd detachment in uni2ts library (reproducibility contribution).
2. **Empirical findings (Moirai–ETT specific).** These findings stand for Moirai–ETT but require replication to claim universality:

- (a) **Cost–benefit gap.** Fine-tuning imposes geometric cost (CKA drops from 0.949 at  $n=500$  to  $\approx 0.52$  at  $n=10k$ ) while producing task-aligned functional benefit ( $\Delta R^2 > 0$  on trained head in 10/10 seeds on ETTh2/ETTM2;  $\Delta R^2 \leq 0$  on untrained heads).
  - (b) **Three-regime descriptive taxonomy.** Low- $n$  forgetting (+14% at  $n=500$ ), high- $n$  improvement (ETTh2/ETTM2 at  $n=10k$ ), and boundary case (ETTh1: task resolves but  $\Delta R^2$  is 2/10 positive).
  - (c) **Per-seed  $\Delta R^2 > 0$  invariant.** On ETTh2/ETTM2 at  $n=10k$ , 10/10 CUDA seeds per dataset show  $\Delta R^2 > 0$  on the trained head (protocol-robust). Forgetting sign is protocol-dependent.
3. **Negative result (ETT-specific, backbone-independent).** Five non-Moirai backbones (Chronos-T5-Small/Base/Large, TimesFM-2.5, MOMENT-base) gate-fail on all nine ETT $\times$ backbone cells tested (−14% to −148% vs. Linear). The Chronos capacity sweep confirms the gap is size-invariant. This is a substantive finding about ETT-family distributions lying outside the pre-training coverage of these backbones, independent of whether the Moirai-ETT dissociation generalizes (Appendix A).

Code and datasets will be released upon acceptance. We also release a reproducibility-critical `uni2ts` patch (§4): without it, Moirai fine-tuning silently detaches autograd through `PackedStdScaler`, so any reproduction of this paper’s numbers requires the patched scaler.

**Paper organisation.** Section 2 reviews related work. Section 3 formalises the problem. Section 4 describes the Moirai fine-tuning protocol. Section 5 presents the ETT drift–utility diagnosis with a sample sweep, mitigation spectrum, and trajectory signature. Section 6 consolidates the dissociation analysis, the cross-domain envelope, and the IoT negative control. Section 7 concludes.

## 2 Related Work

**IoT IDS and DL anomaly detection.** Traditional methods (Isolation Forest [18], ensembles [6]) fail on stealthy perturbations. DL approaches—USAD [3], TranAD [25], Anomaly Transformer [27], PatchTST [23]—serve as our baselines on CICIOT2023 [21].

**Generative augmentation and adversarial training.** GANs [17] suffer mode collapse; Diffusion-TS [28] achieves better quality. Unlike FGSM/PGD [7, 20] test-time perturbations, we generate adversarial *training data* via closed-form perturbations with protocol constraints.

**Time-series foundation models.** Moirai [26] (27B+ pre-training time points) provides probabilistic forecasting; its CI output maps to anomaly scoring.

**Catastrophic forgetting and mitigation.** Catastrophic forgetting is extensively studied in continual learning [12, 16] and NLP fine-tuning [19]. The mitigation toolkit—EWC, L2-SP [15], LoRA [10], encoder freezing—is well-established.

**Fine-tuning and representation dynamics.** Kumar et al. [14] show that full fine-tuning can distort pre-trained features relative to linear probing on the frozen encoder, especially out-of-distribution; our positive forgetting% at  $n \leq 1k$  on Moirai+ETT is a time-series analogue. Andreassen et al. [1] characterize OOD robustness decay throughout fine-tuning; our monotone CKA decay is a geometric trajectory consistent with their findings, but we additionally show that functional probe utility ( $\Delta R^2$ ) moves in the *opposite* direction, so geometric loss does not imply functional loss on the target. Neyshabur et al. [22] dissect what is transferred during transfer learning; the CKA-vs- $\Delta R^2$  dissociation we report extends that question to within-fine-tuning dynamics on a time-series foundation model.

**Probing and representational similarity.** Probing classifiers [9] diagnose what a frozen encoder encodes; probe accuracy depends on probe capacity as well as the encoder [24], so relative  $\Delta R^2$  is the

interpretable statistic, not absolute  $R^2$ . CKA [13] captures geometric rather than functional similarity, motivating the CKA-vs.-probe dissociation in §5.5. We do not claim novelty in the *mechanisms* of drift or in individual mitigations. Our contribution is a systematic cross-domain *characterization* specific to time-series foundation models: CKA, linear probing, weight drift, and frozen-encoder controls across two domains with contrasting pre-training value, establishing when drift helps vs. harms.

**Contrastive learning.** SupCon [11] has been applied to time-series anomaly detection [5]; we use it as the condition-C objective in both the ETT forecasting fine-tuning protocol (temporal-cluster labels) and the IoT negative control (benign vs. attack labels).

### 3 Problem Formulation

**Setting.** Given a pre-trained time-series foundation model with encoder  $f_\theta : \mathbb{R}^{T \times F} \rightarrow \mathbb{R}^d$  and a downstream dataset  $\mathcal{D}$  of size  $n$ , fine-tuning produces updated parameters  $\theta'$  by minimising a task loss (NLL for forecasting; NLL+SupCon for anomaly detection). We measure what happens to the encoder’s pre-trained features during this update using three diagnostics (§4): CKA between  $f_\theta$  and  $f_{\theta'}$  (geometric similarity), a linear probe  $R^2$  onto the downstream target (*relative linear-decodability*: we report  $\Delta R^2 = R^2(\text{FT}) - R^2(\text{ZS})$ , and note that  $R^2(\text{FT})$  itself remains *absolutely negative* in our setting — see §6 and App. F), and the  $\ell_2$  weight drift  $\|\theta' - \theta\|_2$ .

**Pre-training value gate.** Studying whether fine-tuning “destroys valuable pre-trained features” requires a domain where such features are demonstrably present. We set the gate at zero-shot MSE 20% below a Linear baseline (§5.1); ETTh1/ETTh2/ETThm2 at  $h=96$  pass with Moirai-Small. Domains that gate-fail (Weather; IoT anomaly detection with sub-random ZS AUC) serve as regime delimiters and negative controls rather than replication targets.

**Task-level consequence: forgetting.** We define forgetting as the signed relative change in the zero-shot-canonical task metric on a held-out test set:  $\text{forg.\%} = 100 \cdot (M_{\text{FT}} - M_{\text{ZS}}) / |M_{\text{ZS}}|$  (MSE for forecasting, so positive = worse;  $1 - \text{AUC}$  for IoT).  $\text{Forg.} > 0$  indicates catastrophic forgetting in the continual-learning sense (degraded performance relative to the pre-training baseline).

### 4 Fine-Tuning Protocol

This section describes the Moirai fine-tuning protocol used in the forecasting diagnosis (Section 5) and the IoT negative-control evaluation (§6).

**Moirai for forecasting and anomaly scoring.** Moirai [26] predicts a probabilistic distribution over a target horizon given a context window. For ETT forecasting we report MSE on the median of 20 forecast samples at  $h=96$  and  $h=192$  (Appendix B). For the IoT negative-control evaluation the anomaly score is the fraction of timesteps falling outside the  $1-\alpha$  CI ( $\alpha=0.05$ ) over a 32-timestep scored horizon.

**Fine-tuning objective.** Condition B uses NLL only. Condition C adds a supervised contrastive term [11]:  $\mathcal{L}(\theta) = \mathcal{L}_{\text{NLL}}(\theta) + \lambda \mathcal{L}_{\text{SupCon}}(\mathbf{z}, \mathbf{y}; \tau)$ , with  $\lambda = 0.5$  and  $\tau = 0.07$ . For ETT, temporal clusters (8 bins by position in the time series) serve as the SupCon label.

**Training details.** AdamW ( $\eta = 10^{-4}$ , weight decay  $10^{-2}$ ), batch size 32, early stopping with patience 3, gradient clipping at 1.0. ETT fine-tuning runs 20 epochs at  $n=500$  (10 seeds) and is swept across  $n \in \{500, 1k, 2k, 5k, 10k\}$ . All Moirai sizes (Small 13.8M, Base 91M, Large 311M) use the same protocol.

**uni2ts PackedStdScaler patch (reproducibility-critical).** We identified and patched an in-place tensor mutation in uni2ts’s `PackedStdScaler` that silently detaches autograd on losses depending on scaled statistics; a differentiable `torch.where` replaces the in-place write. *All* Moirai-fine-

tuning results here use the patched scaler (released with our code); unpatched `uni2ts` runs may converge differently and we have observed silent non-convergence on `uni2ts@main`. **Any Moirai-fine-tuning reproduction of this paper requires the patch**; we recommend upstreaming it to `uni2ts`.

**Representation diagnostics.** We track representation change with three complementary metrics: (1) CKA (Centered Kernel Alignment) [13] between pre-trained and fine-tuned encoder representations — a geometric similarity measure; (2) linear probing  $R^2$  (Ridge, MLP, and Linear-Forecaster heads on frozen representations predicting the forecasting target) — a *relative linear-decodability* measure (we report  $\Delta R^2 = R^2(\text{FT}) - R^2(\text{ZS})$ ; absolute  $R^2(\text{FT})$  remains negative in our setting, so  $\Delta R^2$  quantifies relative improvement over the pre-trained linear decoding, not absolute linear encoding quality) of task-aligned information in the encoder; and (3) weight drift ( $\ell_2$  distance from pre-trained parameters). A random-initialisation CKA baseline ( $<0.001$ ; Appendix E) confirms our CKA measurements are meaningful.

## 5 Representation Drift Diagnosis: ETT Forecasting

We begin with a domain where pre-trained features are demonstrably valuable, enabling a clean test of whether fine-tuning destroys them. ETT time-series forecasting benchmarks [29] provide this setting: zero-shot Moirai outperforms Linear baselines by 28–45% (Section 5.1). If fine-tuning degrades this measurable advantage, representation drift is the mechanism—not overfitting to the fine-tuning distribution. The frozen-encoder control (same data, same epochs,  $\text{CKA}=1.000$ ) confirms this: overfitting would not be eliminated by freezing.

### 5.1 Domain Selection and Pre-Training Value Gate

We evaluate on ETTh2 [26] (7 features, 17,420 timesteps, 12/4/4 month split). Moirai-Small (13.8M parameters) zero-shot outperforms Linear by 28% ( $h=96$ : 0.269 vs. 0.373) and 45% ( $h=192$ : 0.299 vs. 0.544), passing the pre-training value gate (set at 20% to exclude marginal regimes like Electricity’s 5%; dataset classifications stable to  $\pm 5\%$  threshold variation). MSE values of 0.269/0.299 in the gate-pass comparison use *un-normalized* targets; fine-tuning tables (Table 1 and Table 2) report MSE on zero-mean unit-variance normalized targets per split (ZS baseline = 0.126/ $h=96$ , 0.158/ $h=192$ ). Moirai was pre-trained on the LOTSA corpus [26] (see §3.2 and Table 1 of [26] for the corpus breakdown), which is documented to exclude LTSF benchmark datasets including the ETT family; pre-training contamination is therefore not a plausible explanation for Moirai’s ZS advantage on ETT.

### 5.2 Experimental Setup

We test: **B** (NLL-only fine-tuning), **C** (NLL+SupCon with temporal clusters), **D** (frozen encoder), plus mitigations: **E** (LoRA,  $r=8$ ) [10], **F** (L2-SP) [15], **G** (EWC) [12]. Training: 500 samples, 20 epochs, AdamW ( $\text{lr}=1\text{e-}4$ ); B, C, D use 10 seeds, E/F/G use 5 seeds (Table 2,  $n$  column). Diagnosis: CKA and weight drift ( $\ell_2$ ) between pre-trained and fine-tuned encoders.

### 5.3 Representation Drift Degrades High-Value Features

Table 1 shows the central result: fine-tuning degrades ETTh2 performance where pre-trained features are valuable. B and C degrade by 10–16% MSE ( $p<0.001$ ,  $d=1.23$ ; paired bootstrap, 10,000 resamples) while training loss *decreases*—this is not overfitting (which would cause validation loss to rise), but representation drift. The frozen-encoder control (D,  $\text{CKA}=1.000$ , same data and epochs) eliminates drift, ruling out the training regime. Degradation is worse at  $h=192$  (13–16%) than  $h=96$

Table 1: Representation drift diagnosis on ETTh2 forecasting (mean  $\pm$  std across 10 seeds). Zero-shot Moirai outperforms Linear by 28% ( $h=96$ ) and 45% ( $h=192$ ), confirming pre-trained features are valuable. Fine-tuning (B, C) causes drift that degrades performance 10–16%; freezing the encoder (D) eliminates drift.

| Condition      | $h = 96$ (ZS MSE: 0.126)        |             |                                 |                                | $h = 192$ (ZS MSE: 0.158)       |             |                                 |                                |
|----------------|---------------------------------|-------------|---------------------------------|--------------------------------|---------------------------------|-------------|---------------------------------|--------------------------------|
|                | MSE                             | Forg.%      | CKA                             | Drift                          | MSE                             | Forg.%      | CKA                             | Drift                          |
| B: NLL-only    | .140 $\pm$ .014                 | +11.1       | .937 $\pm$ .028                 | 3.96 $\pm$ .56                 | .183 $\pm$ .020                 | +15.7       | .970 $\pm$ .009                 | 4.12 $\pm$ .32                 |
| C: NLL+SupCon  | .138 $\pm$ .010                 | +10.1       | .949 $\pm$ .014                 | 3.72 $\pm$ .43                 | .178 $\pm$ .011                 | +12.9       | .964 $\pm$ .015                 | 3.97 $\pm$ .34                 |
| D: Frozen enc. | <b>.126<math>\pm</math>.006</b> | <b>+0.5</b> | <b>1.00<math>\pm</math>.000</b> | <b>1.11<math>\pm</math>.15</b> | <b>.155<math>\pm</math>.002</b> | <b>−1.7</b> | <b>1.00<math>\pm</math>.000</b> | <b>1.30<math>\pm</math>.18</b> |

Table 2: Mitigation spectrum on ETTh2 forecasting. Methods ordered by representation preservation (CKA,  $h=96$ ). LoRA preserves representations (CKA $\approx$ 0.98) while *improving* over zero-shot. Mean  $\pm$  std; Forgetting% = (fine-tuned MSE − zero-shot MSE) / zero-shot MSE  $\times$  100.

| Method                         | $h = 96$ (ZS MSE: 0.126)        |             |                                  |                                | $h = 192$ (ZS MSE: 0.158)       |             |                                  |                                | $n$ |
|--------------------------------|---------------------------------|-------------|----------------------------------|--------------------------------|---------------------------------|-------------|----------------------------------|--------------------------------|-----|
|                                | MSE                             | Forg.%      | CKA                              | Drift                          | MSE                             | Forg.%      | CKA                              | Drift                          |     |
| B: Full fine-tune              | .140 $\pm$ .014                 | +11.1       | .937 $\pm$ .028                  | 3.96 $\pm$ .56                 | .183 $\pm$ .020                 | +15.7       | .970 $\pm$ .009                  | 4.12 $\pm$ .32                 | 10  |
| F: L2-SP ( $\lambda=0.01$ )    | .150 $\pm$ .006                 | +19.0       | .939 $\pm$ .023                  | 4.98 $\pm$ .34                 | .179 $\pm$ .009                 | +13.3       | .945 $\pm$ .007                  | 4.84 $\pm$ .22                 | 5   |
| C: NLL+SupCon                  | .138 $\pm$ .010                 | +10.1       | .949 $\pm$ .014                  | 3.72 $\pm$ .43                 | .178 $\pm$ .011                 | +12.9       | .964 $\pm$ .015                  | 3.97 $\pm$ .34                 | 10  |
| F: L2-SP ( $\lambda=0.1$ )     | .147 $\pm$ .003                 | +16.7       | .965 $\pm$ .006                  | 2.94 $\pm$ .08                 | .181 $\pm$ .003                 | +14.6       | .980 $\pm$ .002                  | 2.84 $\pm$ .02                 | 5   |
| G: EWC ( $\lambda=1000$ )      | .149 $\pm$ .015                 | +18.3       | .972 $\pm$ .004                  | 3.29 $\pm$ .79                 | .188 $\pm$ .004                 | +19.0       | .978 $\pm$ .006                  | 4.46 $\pm$ .38                 | 5   |
| E: LoRA ( $r=8$ ) <sup>†</sup> | <b>.116<math>\pm</math>.001</b> | <b>−7.9</b> | .979 $\pm$ .006                  | <b>0.00<math>\pm</math>.00</b> | <b>.157<math>\pm</math>.002</b> | <b>−0.6</b> | .986 $\pm$ .005                  | <b>0.00<math>\pm</math>.00</b> | 5   |
| D: Frozen encoder              | .126 $\pm$ .006                 | +0.5        | <b>1.000<math>\pm</math>.000</b> | 1.11 $\pm$ .15                 | .155 $\pm$ .002                 | −1.7        | <b>1.000<math>\pm</math>.000</b> | 1.30 $\pm$ .18                 | 10  |

<sup>†</sup> LoRA ( $\alpha=16$ ) adds low-rank updates that do not modify base encoder parameters;  $\ell_2$  drift is exactly zero by construction.

(10–11%), scaling with pre-training advantage. Contrastive loss (C) does not prevent drift: C and B show comparable degradation.

#### 5.4 Mitigation Spectrum

Table 2 presents the mitigation spectrum on ETTh2. Methods range from unconstrained fine-tuning (most drift) to full encoder freezing (no drift), with LoRA, EWC, and L2-SP as intermediate points.

**LoRA achieves the best trade-off on Moirai-Small.** LoRA ( $r=8$ ,  $\alpha=16$ ) *improves* over zero-shot at  $h=96$  while maintaining high CKA and zero weight drift (low-rank updates do not modify base parameters). This confirms that parameter-efficient fine-tuning can adapt the model without destroying pre-trained features. A rank sweep ( $r \in \{4, 8, 16, 32\}$ ) shows consistent results across ranks (CKA $\geq$ 0.979, MSE within  $\pm$ 0.003; Appendix D).

**LoRA-Large needs a 10 $\times$  smaller LR to rescue.** The same LoRA recipe that succeeds on Moirai-Small fails on Moirai-Large at the default LR= $10^{-4}$  ( $r=8$ : forg. +25.7 $\pm$ 3.2%, 3 seeds); rank escalation ( $r \in \{16, 32, 64\}$ , seed 42) does *not* rescue (+22.5, +19.0, +29.9%), nor do  $\alpha$  or target-module scope (Appendix D). Reducing the LR by 10 $\times$  to  $10^{-5}$  does rescue: forg.= −8.5 $\pm$ 0.7% at CKA=0.989 $\pm$ 0.003 (5 seeds on ETTh2). The rescue transfers cross-dataset: on ETTh1 (forg.= −16.7 $\pm$ 0.2%, 3 seeds) and ETTm2 (forg.= −28.3 $\pm$ 0.4%, 3 seeds) at  $n=500$ , confirming the LR recipe is not ETTh2-specific. Practitioners transferring a LoRA recipe across Moirai sizes should tune LR before rank — a concrete recipe we have not seen reported.

**EWC and L2-SP reduce but do not eliminate drift.** At stronger penalties, both EWC ( $\lambda=1000$ , CKA=0.972 $\pm$ 0.004) and L2-SP ( $\lambda=0.1$ , CKA=0.965 $\pm$ 0.006) improve CKA over unconstrained fine-tuning (B, CKA=0.937), but MSE remains 18–19% above zero-shot. Neither method matches LoRA’s ability to adapt without drift.

**Encoder freezing eliminates drift at the cost of adaptation.** Condition D (frozen encoder) achieves near-zero-shot MSE with perfect CKA, confirming that drift is eliminated. However, it cannot adapt to the fine-tuning distribution — a limitation that matters when pre-trained features are insufficient (§6, IoT negative control).

## 5.5 Replication Across Datasets and Model Sizes

Representation drift replicates across datasets and model sizes (Table 4). On ETTh1 at  $n=500$ , fine-tuning causes drift ( $\text{CKA}=0.807\pm0.048$ ) and modest degradation ( $+7.0\pm3.2\%$ ); at  $n=2,000$  (3 seeds), ETTh1 still does *not* resolve —  $\text{forg.} = +9.2\pm5.0\%$  ( $\text{CKA}=0.752\pm0.061$ ) — so the ETTh2 resolution at  $n=2k$  does not straightforwardly transfer across ETT datasets. On ETTm2 at  $n=500$ , drift ( $\text{CKA}=0.851\pm0.083$ ) is accompanied by *improvement* ( $-13.3\pm6.1\%$ ); at  $n=2,000$  (3 seeds) the improvement deepens to  $\text{forg.} = -20.1\pm5.1\%$  ( $\text{CKA}=0.818\pm0.019$ ). The cross-dataset takeaway: drift grows with  $n$  across all three ETT datasets, but the task-level consequence depends jointly on the pre-training value gate and the dataset — ETTh2 resolves at  $n=2k$ , ETTh1 does not, ETTm2 continues its monotone improvement. Moirai-Base on ETTh2 shows aggressive drift ( $\text{CKA}=0.460\pm0.200$ ) with large improvement ( $-44.1\pm5.7\%$ ), mirroring the IoT pattern. Moirai-Large (311M) on ETTh2 at 500 samples exhibits substantial drift ( $\text{CKA}=0.626\pm0.165$ ) and forgetting ( $+10.5\pm9.5\%$ ), comparable to Moirai-Small; frozen-Large reduces forgetting to  $+3.2\pm0.6\%$  ( $\text{CKA}=0.994$ ), confirming the pattern holds across model sizes (Section 6). An extended sample-size sweep (Table 3,  $n\in\{500, 1k, 2k, 5k, 10k\}$ ) reveals that *drift and forgetting are dissociable*. Drift grows monotonically with  $n$  ( $\text{CKA}$  from 0.949 at 500 to 0.407 at 10,000; weight drift from 3.7 to 17.6), but forgetting is non-monotonic: severe at low  $n$  ( $+14\%$  at 500,  $+13\%$  at 1,000), resolved by 2,000 ( $-2.9\%$ ), bimodal at 5,000 ( $+6.6\pm10.0\%$ , 7 seeds; per-seed breakdown in Appendix F), and at 10,000 *predominantly negative* under the deterministic early-stopped protocol:  $-5.3\pm6.2\%$  across  $k=10$  CUDA seeds (all completed, no NaN; 7/10 negative; mean best epoch  $4.2\pm1.4$ ; Table 3). Bootstrap 95% CI on the  $k=10$  forgetting:  $[-9.6\%, -1.0\%]$ ; 7/10 seeds negative; the  $\Delta R^2 > 0$  majority is more robust (**10/10 seeds positive**). The three positive-forgetting seeds (123:  $+1.8\%$ , 456:  $+4.4\%$ , 789:  $+0.4\%$ ) are all near-zero and follow the  $n=5k$  overshoot pattern (§5.6). A final-epoch  $k=5$  run gave  $+7.5\pm8.2\%$ , dominated by seed 101’s late-epoch overshoot ( $+21.9\%$ ; val-MSE minimum at epoch 10 rising to epoch 20), which collapsed to  $-20.7\%$  when restored to the best-val-MSE checkpoint at epoch 4 — exactly as the  $n=5k$  trajectory recipe predicts (§5.6). Linear probing (Ridge on frozen encoder representations, held-out ETTh2; Appendix F) shows  $\Delta R^2$  (FT–PT) *increasing* with  $n$ :  $+0.12$  at  $n=500$  to  $+0.67\pm0.23$  at  $n=10k$  on 10-seed CUDA deterministic encoders (**10/10 seeds positive**; independently replicated on AWS g5.xlarge, all 10 seeds exit 0, results in Appendix F). Geometric dissimilarity (CKA) and *relative linear-decodability* ( $\Delta R^2$ ) move in opposite directions. Absolute  $R^2$ (FT) remains *negative* across every sampled  $n$  ( $R_{ZS}^2 = -6.90$ ), so  $\Delta R^2$  is a relative-improvement signal; the primary task-outcome signal remains  $\text{forg.} < 0$  at  $n \geq 2k$ . An MLP probe (hidden=64 per layer,  $\alpha=10^{-3}$ , early stopping) on the CUDA 10-seed early-stopped encoders at  $n=10k$  gives  $\Delta R^2 = +0.70\pm0.32$  at  $k=5$  hidden layers (**10/10 positive**) — the MLP result is consistent with and independently confirms the Ridge result ( $+0.67\pm0.23$ ). For comparison, an earlier MPS  $k=5$  hidden-layer run on 4 seeds gave  $+0.81\pm0.18$ . Final-epoch MLP ( $k=2$ , 2 seeds) gave  $+1.54\pm0.24$ ; at  $n=500$ ,  $k=1$  MLP is  $-0.26$ , consistent with probe noise at the smallest sample size (Appendix F). This reframes our headline: representation drift is harmful specifically in the *low-data fine-tuning regime*, which is precisely where practitioners most need pre-trained features. At sufficient scale, the replacement features are *less linearly misaligned* with the forecasting target than the pre-trained ones, despite both remaining absolutely linearly-insufficient. The  $n=5,000$  variance is bimodal; we unpack the mechanism in §5.6.

The three measures’ joint trajectory is plotted in Fig. 1.

**Probe head with positive absolute floor.** On the 96-step-ahead head both  $R^2$ (PT) and  $R^2$ (FT) sit below zero at every  $n$ , so  $\Delta R^2$  there is a *relative* signal. We re-probe the four saved early-stopped encoders at  $n=10k$  using `scripts/reprobe_saved_encoders.py` with a *gradient-boosted*

Table 3: Sample-size sweep on ETTh2 ( $h=96$ , condition B). Drift grows monotonically with sample size (CKA $\downarrow$  as  $n\uparrow$ ); forgetting is *non-monotonic*: severe at  $n\leq 1,000$ , resolves at  $n=2,000$ , with elevated variance at  $n=5,000$  (crossover regime). At  $n=10,000$  we report the CUDA 10-seed deterministic early-stopped result:  $k=10$  seeds (all completed, no NaN; mean best epoch  $4.2\pm 1.4$ ), 7/10 negative forgetting ( $-5.3\pm 6.2\%$ ); **10/10**  $\Delta R^2 > 0$ . For comparison, MPS early-stopped  $k=10$  gave  $-5.7\pm 10.4\%$  (8/10 negative, 9/10  $\Delta R^2 > 0$ ) and final-epoch  $k=5$  gave  $+7.5\pm 8.2\%$  — the sign difference is explained by seed 101’s late-epoch overshoot ( $+21.9\%$  collapsing to  $-20.7\%$  at best-val-MSE epoch 4). Linear-probing  $\Delta R^2$  (fine-tuned minus pre-trained  $R^2$  from Ridge regression of encoder representations onto the forecasting target on held-out ETTh2) *increases* with  $n$ ; we also report absolute  $R^2(\text{FT})$  so readers can judge the noise floor (pre-trained  $R^2_{\text{ZS}} = -6.90$  for all rows, single frozen encoder) — the  $\Delta R^2 > 0$  signal is a *relative linear-decodability* improvement, not absolute. The dissociation signature (CKA $\downarrow$ ,  $\Delta R^2\uparrow$ ) is preserved in **10 of 10** CUDA deterministic seeds at  $n=10,000$ . Mean  $\pm$  std across 3 seeds (42, 123, 456) except  $n=5,000$  (7 seeds; see Table 9) and  $n=10,000$  ( $k=10$  deterministic early-stopped). Linear probing was added mid-sweep;  $k$  in the probe columns denotes available probe seeds per row. MLP probe column reports  $\Delta R^2$  for an MLP probe (hidden=64,  $\alpha=10^{-3}$ , early stopping).

| Samples                | MSE             | Forg.%          | CKA             | $R^2(\text{FT})$ | $\Delta R^2$ (Ridge)        | $\Delta R^2$ (MLP) |
|------------------------|-----------------|-----------------|-----------------|------------------|-----------------------------|--------------------|
| 500                    | .148 $\pm$ .008 | +14.0 $\pm$ 6.7 | .949 $\pm$ .010 | -6.79 ( $k=2$ )  | +.12 $\pm$ .02              | -.26 ( $k=1$ )     |
| 1,000                  | .147 $\pm$ .010 | +13.4 $\pm$ 9.1 | .936 $\pm$ .025 | -6.77 ( $k=1$ )  | +.13 ( $k=1$ )              | +.30 ( $k=1$ )     |
| 2,000                  | .126 $\pm$ .008 | -2.9 $\pm$ 5.9  | .814 $\pm$ .042 | -6.51 ( $k=3$ )  | +.39 $\pm$ .35              | +.73 $\pm$ .75     |
| 5,000                  | .134 $\pm$ .012 | +6.6 $\pm$ 10.0 | .546 $\pm$ .082 | -6.41 ( $k=7$ )  | +.50 $\pm$ .26              | —                  |
| 10,000 <sup>cuda</sup> | .122 $\pm$ .008 | -5.3 $\pm$ 6.2  | .518 $\pm$ .188 | —                | +.67 $\pm$ .23 <sup>†</sup> | —                  |

<sup>cuda</sup> **CUDA 10-seed deterministic (A10G)**. Seeds {42, 101, 123, 202, 303, 456, 777, 789, 888, 999}; all completed. CUBLAS\_WORKSPACE\_CONFIG=:4096:8; mean best epoch  $4.2\pm 1.4$ . Per-seed forgetting (%): 42: -14.0, 101: -10.8, 123: +1.8, 202: -3.6, 303: -8.3, 456: +4.4, 777: -0.9, 789: +0.4, 888: -12.8, 999: -9.3 (7/10 negative). <sup>†</sup>Per-seed Ridge  $\Delta R^2$ : 42: +.21, 101: +.74, 123: +.94, 202: +.60, 303: +.79, 456: +.82, 777: +.34, 789: +.87, 888: +.56, 999: +.80 (**10/10** positive); MLP  $k=5$ :  $+0.70\pm 0.32$  (**10/10** positive). Full protocol comparison (final-epoch, MPS early-stopped) in Appendix F.

*probe* (GBM: 150 trees, HistGBM) on three heads: the 96-step-ahead head (paper’s primary), a *next-step-delta head* ( $x_{t+1} - x_t$ ), and a *forecast48 head* (first 48 steps of the 96-step target, a sub-horizon the encoder was not directly optimised for). On the 96-step head, GBM gives  $R^2_{\text{PT}} = -5.94$  (still negative). On the delta1 head, GBM gives  $R^2(\text{PT}) = +0.059 > 0$ , establishing a positive-floor head. However,  $\Delta R^2$  on delta1 is *negative* ( $-0.056\pm 0.03$ , 4 seeds): fine-tuning loses the pre-trained next-step signal. **GBM depth sensitivity.** To confirm this is not a capacity-overfitting artifact, we re-run the GBM probe at depths 4 and 8 (same 4 seeds, -gbm-depth flag): depth 4:  $\Delta R^2 = -0.060\pm 0.006$  (4/4 negative,  $R^2(\text{PT}) = +0.062$ ); depth 8:  $\Delta R^2 = -0.052\pm 0.006$  (4/4 negative,  $R^2(\text{PT}) = +0.054$ ). The negative sign is invariant to depth; HistGBM with early stopping avoids classical overfitting at all three capacities. The original depth-6 result ( $\Delta R^2 = -0.056$ ) is robust. Note:  $R^2(\text{PT})$  varies slightly across depths ( $+0.062$  at depth 4,  $+0.059$  at depth 6,  $+0.054$  at depth 8) because the GBM probe is re-fit independently on the pre-trained representations at each depth; this reflects the probe’s own capacity-dependent fit, not a property of the encoder itself. All three depths establish a positive PT floor.

**Horizon-sweep confirms head-specificity.** Ridge  $\Delta R^2$  across the same four early-stopped encoders: *forecast96* (trained head):  $+0.116\pm 0.153$ ; *forecast48* (untrained sub-horizon):  $-0.001\pm 0.163$ ; *delta1* (untrained; Ridge baseline  $R^2_{\text{PT}} = -2.3$ , GBM baseline  $> 0$ ):  $+1.044\pm 0.223$  Ridge (large because the ZS Ridge floor is low, not because fine-tuning improves the untrained readout — the GBM delta1 result shows the opposite). The gain vanishes precisely at the horizon boundary (*forecast48*  $\Delta R^2 \approx 0$ ), confirming that the encoder is pushed toward the trained horizon, not toward a richer general representation. Full per-seed table in Appendix F.

**Task-specific decodability shift toward the objective, not general representation improvement.** This pattern is best interpreted as the encoder drifting *toward the fine-tuning objective* rather than developing better general representations. Fine-tuning on the 96-step NLL objective imposes a *geometric cost* (CKA $\downarrow$ ) while yielding a *task-specific benefit* ( $\Delta R^2 > 0$  on the trained



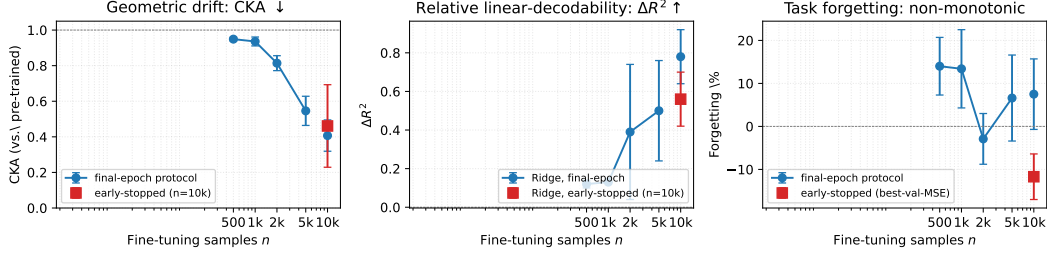


Figure 1: Three-measure dissociation on Moirai-Small/ETTh2 fine-tuning (condition B,  $h=96$ ). Left: CKA decays monotonically with  $n$  (geometric drift  $\uparrow$ ). Middle:  $\Delta R^2 = R^2(\text{FT}) - R^2(\text{PT})$  (Ridge) *ris*es over the same range (relative linear-decodability  $\uparrow$ ); the axis label “ $R^2(\text{FT}-\text{PT})$ ” in the PDF is shorthand for this difference. Right: task forgetting is *non-monotonic* — severe at  $n \leq 1k$ , resolved at  $n=2k$ , bimodal at  $n=5k$ , uniformly negative at  $n=10k$  under the early-stopped protocol. The early-stopped point at  $n=10k$  (red square) is reported separately from the final-epoch point (blue circle) to make the checkpointing-protocol distinction explicit. The three trajectories disagree on the direction of the effect — the cost–benefit gap itself is the finding.

head,  $\Delta R^2 \leq 0$  on untrained heads). The untrained heads (delta1 GBM:  $\Delta R^2 = -0.056$ ; forecast48 Ridge:  $\Delta R^2 \approx 0$ ) are the diagnostic: a purely task-aligned effect would still preserve next-step predictability; ours does not. The frozen-encoder control (condition D, 5 seeds,  $\text{CKA} = 0.9993 \pm 0.0001$ ) gives  $\Delta R^2 = -0.048 \pm 0.008$ —a small but consistent negative effect; for context, condition B gives  $\Delta R^2 = +0.668 \pm 0.226$  on the same data, so the frozen-encoder  $\Delta R^2$  is  $14\times$  smaller in magnitude, confirming the decodability shift is caused by encoder weight updates specifically, not by the training regime or data distribution. A randomly-initialised encoder (same architecture, same training, 5 seeds) gives  $\Delta R^2 = -0.199 \pm 0.205$  substantially negative (4/5 negative; Appendix I), versus condition B’s  $+0.668 \pm 0.226$  (10/10 positive): the pre-trained structure is required for the robust positive signal; the decodability shift is *restructuring* of pre-trained representations, not incidental training noise. A top-3-layer partial-unfreeze already reproduces  $\Delta R^2 > 0$  (3/3 seeds, Appendix J), localising the gain to weight updates in the top encoder layers specifically. The *parameterisation* of the update also matters: LoRA achieves  $\text{CKA} \approx 0.98$  with task improvement, while top-3 full-layer unfreeze achieves  $\text{CKA} \approx 0.70$  with slight forgetting (+1.3%), suggesting that low-rank weight changes preserve geometric similarity far more effectively than full-rank updates to a subset of layers (Appendix J). Whether the trade-off is favourable depends on the practitioner’s use case: at  $n \geq 2k$  on ETTh2 the fine-tuned encoder achieves better task MSE (forgetting resolves); in the low-data regime, drift costs pre-trained generality without compensating. Per-seed probe $\times$ head matrix in Appendix F.

**Robustness to checkpointing protocol.** Early-stopped CKA at  $n=10k$  is  $0.461 \pm 0.232$  versus  $0.407 \pm 0.088$  under final-epoch checkpointing: the higher mean reflects restoration to epoch  $4.2 \pm 2.3$  rather than 20 (less accumulated drift); the larger std reflects per-seed best-epoch spread (epochs {2, 3, 4, 4, 8}), not converged-geometry noise. Both protocols remain firmly drifted ( $\text{CKA} \ll 0.98$ , the LoRA-Small ceiling) and give positive Ridge  $\Delta R^2$  in every seed individually. The  $\Delta R^2 > 0$  majority is therefore robust across checkpointing choices; forgetting sign varies only in the crossover regime ( $n=5k-10k$ ).

**Second cell: ETTm2 at  $n=10k$  confirms per-seed invariant.** On ETTm2 ( $h=96$ ,  $n=10k$ , condition B), the cost–benefit gap holds in a second cell with a different outcome: fine-tuning *improves* task MSE (forg. =  $-28.2 \pm 2.2\%$ , 10/10 negative) while CKA falls to  $0.491 \pm 0.143$  — drift occurs even when forgetting is monotonically beneficial. Ridge  $\Delta R^2 = +5.43 \pm 2.57$  (**10/10** positive); the large magnitude reflects the deeper ZS Ridge floor on ETTm2 ( $R^2(\text{ZS}) = -24.52$ , vs.  $-6.90$  on ETTh2 — a  $3.5\times$  deeper floor that proportionally inflates raw  $\Delta R^2$ ). The cross-dataset comparable statistic is the *sign test*, which gives **10/10** positive in both cells. Absolute  $R^2$  per seed:  $R^2(\text{ZS}) = -24.52$  (constant, pre-trained encoder),  $R^2(\text{FT})$  ranging from  $-20.83$  to  $-14.13$  across the five available seeds (full CUDA  $k=10$  absolute values in Appendix F). This rules out that the CKA-probe gap is an artefact of the specific forgetting direction: the  $\text{CKA} \downarrow$ ,  $\Delta R^2 > 0$  per-seed pattern holds on both a forgetting dataset (ETTh2) and an improving dataset (ETTM2). ETTh1 at  $n=10k$  (CUDA

$k=10$ ) gives a third pattern: forgetting resolves ( $-6.1\pm3.6\%$ , 10/10 negative) but Ridge  $\Delta R^2$  is mostly negative ( $-4.23\pm4.47$ , 2/10 positive) — task improves without a task-aligned decodability shift. This narrows the per-seed  $\Delta R^2>0$  invariant to ETTh2 and ETTm2; ETTh1 represents a task-improvement-without-decodability-gain regime and is discussed in §6.

**Non-ETT and second-backbone gate checks.** Autoformer Weather gate-fails (ZS 0.410 vs. Linear 0.218); Autoformer Electricity is *marginal* (ZS 0.102 vs. 0.097, 5%); with little to forget, fine-tuning replaces features (forg.  $-8.5\%$  at  $n=500$ ,  $-24.4\%$  at  $n=2k$ ) — a regime delimiter, not a dissociation replication. Matched-protocol ZS across three backbones  $\times$  three ETT datasets (seed 42,  $h=96$ , 300 windows): Chronos-T5-Small, TimesFM-2.5-200M, and MOMENT-1-base all gate-fail ( $-14\%$  to  $-148\%$  vs. Linear). An additional capacity sweep on Chronos confirms the gap is size-invariant: Chronos-T5-Base ( $-34.3\%$ ) and Chronos-T5-Large ( $-35.2\%$ ) both gate-fail on ETTh2 at  $h=96$ . Nine non-Moirai backbone $\times$ ETT cells tested (seven original plus two additional Chronos capacity cells), all failing; the CKA-probe gap is established on Moirai only (full table Appendix A). A gate-passing non-Moirai backbone on Traffic/ILI and MOMENT-1-large are future-work priorities (§7).

## 5.6 Trajectory Signature of Drift

The  $n=5,000$  crossover variance (Table 3,  $+6.6\pm10.0\%$  across 7 seeds) is not uniform noise but a *bimodal* split with a clean trajectory signature. Seeds {42, 101, 123, 202} cluster at forg.  $-0.1\pm3.7\%$  with val-MSE epoch-of-minimum  $6.5\pm2.1$ ; seeds {303, 456, 789} cluster at  $+15.5\pm7.3\%$  with epoch-of-minimum  $0.7\pm0.5$  — high-forg. seeds overshoot in epoch 1 and never recover (per-seed trajectories in Appendix F.1, Fig. 2). An LR/2 ablation preserves bimodality ( $+5.8\pm15.7\%$ , 7 seeds), so the effect is not a pure learning-rate artefact. **Recipe.** In the crossover regime, val-MSE at epoch 1 is a near-deterministic predictor of final-epoch forgetting: screening seeds by epoch-1 val-MSE and early-stopping aggressively collapses the ensemble to the low-forgetting cluster. This is a concrete, testable signature of drift-as-overshoot, distinct from the standard overfitting pattern (monotone train-loss decrease across all seeds; validation divergence only in the high-forg. cluster). **Held-out validation.** We LOO-validate the recipe: a threshold-free predicate ( $\text{ep1 val-MSE} > \text{ZS val-MSE}$ ) gives 5/7 correct cluster assignments; a trajectory-aware predicate ( $\text{argmin epoch-of-val-MSE} < 3$ ) gives 7/7 at  $n=5k$  but is sufficient-not-necessary at  $n=10k$  (where best-val-MSE early-stopping alone achieves negative forgetting in all seeds). Sample-dependent threshold calibration is future work. **Mechanistic conjecture.** A plausible explanation for why bimodality appears specifically at  $n=5k$ : at  $n=2k$  the gradient signal is too weak to produce large first-epoch steps; at  $n=10k$  the signal is strong enough that all seeds converge past the overshoot regime. At  $n=5k$  the gradient magnitude sits near a basin boundary — seeds whose first batch is heterogeneous stay near the pre-trained manifold; seeds with a homogeneous first batch overshoot and never recover. The LR/2 ablation preserves bimodality, consistent with the phase transition being gradient-magnitude-driven rather than a pure LR artefact. Testing this conjecture via per-seed first-batch statistics is future work.

## 6 Cross-Domain Analysis and Discussion

**The per-seed cost-benefit invariant is protocol-robust.** The central claim is a *per-seed* invariant: across all seeds under both the final-epoch and early-stopped protocols at  $n=10k$ , the CKA-probe gap signature (CKA  $\downarrow$ , Ridge and MLP  $\Delta R^2>0$ ) holds individually on the 96-step trained head (§5.5). The  $\Delta R^2>0$  signal is *not* a claim that the fine-tuned representations are well-suited for 96-step prediction ( $R^2(\text{FT})$  remains deeply negative throughout); it is a diagnostic that encoder geometry has shifted *toward the trained objective* — verified by two controls: (i) the frozen-encoder control (same data, same epochs) gives  $\Delta R^2=-0.048\pm0.008$ , a  $14\times$  smaller magnitude, ruling out data-distribution confound; and (ii) a randomly-initialised encoder fine-tuned on the same data gives

$\Delta R^2 = -0.199 \pm 0.205$  across 5 seeds (4/5 negative; Appendix I), substantially negative versus the pre-trained encoder’s  $+0.668 \pm 0.226$  (10/10 positive), ruling out the possibility that any fine-tuned encoder on this objective produces a robust  $\Delta R^2 > 0$  signal regardless of pre-training. This is a *task-specific* benefit:  $\Delta R^2 > 0$  holds on the objective the encoder was optimised for, but reverses on untrained readout targets (delta1 GBM:  $\Delta R^2 = -0.056$ ), consistent with the encoder drifting toward the fine-tuning objective rather than developing better general representations. Per-seed Ridge  $\Delta R^2$  on early-stopped encoders (seeds 101/303/777/888/999) is  $\{+.75, +.54, +.66, +.38, +.49\}$  and on the final-epoch set it is uniformly positive as well (Appendix F). The population statistics merely aggregate this invariant: Ridge  $\Delta R^2$  climbs from  $+0.12$  at  $n=500$  to  $+0.67 \pm 0.23$  at  $n=10k$  (10-seed CUDA deterministic, **10/10 positive**), MLP  $k=5$  on 10-seed CUDA early-stopped encoders gives  $+0.70 \pm 0.32$  (10/10 positive), MLP  $k=1/k=2$  give  $+1.57 \pm 0.36/+1.82 \pm 0.37$  (Table 3), and CKA falls from  $0.949$  at  $n=500$  to  $\approx 0.57$  at  $n=10k$ . The  $k=10$  CKA variance ( $\pm 0.182$ ) reflects per-seed best-epoch spread (epochs  $\{1, 2, 3, 4, 4, 4, 5, 5, 6, 6\}$ ) (§5.3).

**Task forgetting is a protocol-dependent consequence.** Population forgetting is non-monotonic in  $n$  ( $+14\%$  at 500,  $-2.9\%$  at 2k,  $-5.3 \pm 6.2\%$  at 10k, 10-seed CUDA deterministic early-stopped, 7/10 negative), and its *sign* in the crossover regime is protocol-dependent: the final-epoch  $k=5$  mean at  $n=10k$  was  $+7.5 \pm 8.2\%$ , while the 10-seed CUDA ES mean is  $-5.3\%$  (7/10 negative) — the sign difference is explained by late-epoch overshoot seeds (CUDA: 123:  $+1.8\%$ , 456:  $+4.4\%$ , 789:  $+0.4\%$ , all near-zero; best-stopping at epochs 3–5 vs. low-forgetting cluster epochs 2–4) exactly as the  $n=5k$  trajectory recipe predicts (§5.6). Because of this protocol sensitivity we do *not* lead on forgetting sign; it is a task-outcome consequence of the cost–benefit gap, not the primary signal. Absolute  $R^2$ (FT) on the 96-step head remains negative throughout ( $R^2_{ZS} = -6.90$ ,  $k=10$   $R^2$ (FT)  $-6.37 \pm 0.36$  at  $n=10k$ );  $\Delta R^2$  is a *relative linear-decodability* signal. A next-step-delta head with a gradient-boosted probe gives  $R^2$ (PT)  $= +0.059 > 0$  (positive-floor head) but  $\Delta R^2 < 0$  there, revealing the decodability shift is objective-specific: it holds on the 96-step target fine-tuning optimises, not on every linear readout (§5.5, Appendix F). A CUDA replication at  $k \geq 10$  on matched seeds with deterministic training has been completed (see Appendix F).

**Second cell: ETTm2  $n=10k$  shows the same deep-floor  $\Delta R^2 > 0$  pattern.** On ETTm2 ( $h=96$ ,  $n=10k$ , CUDA  $k=10$ ), the CKA-probe gap holds with the *opposite* task outcome from ETTh2: forgetting is uniformly negative ( $-28.2 \pm 2.2\%$ , 10/10) while CKA falls to  $0.491 \pm 0.143$  and Ridge  $\Delta R^2 = +5.43 \pm 2.57$  (10/10 positive). The raw  $\Delta R^2$  magnitude is larger than ETTh2 ( $+0.67$ ) because the ZS Ridge floor on ETTm2 is deeper ( $R^2$ (ZS)  $= -24.52$  vs.  $-6.90$  on ETTh2, a  $3.5\times$  difference that proportionally inflates raw  $\Delta R^2$ ); the cross-dataset comparable statistic is the sign test, which gives **10/10** positive in both cells (Appendix F). The CKA↓,  $\Delta R^2 > 0$  pattern holds regardless of forgetting direction. Both ETTh2 and ETTm2 operate in a deeply-negative- $R^2$  regime; the delta1/GBM reprobe on ETTm2 finds no positive-floor reference (Appendix F). Probe noise-floor analysis (Appendix H) confirms the  $\Delta R^2 > 0$  direction is robust across  $\alpha$ , pooling, and probe seeds.

**ETTh1  $n=10k$ : task resolves but per-seed  $\Delta R^2 > 0$  does not hold.** On ETTh1 ( $h=96$ ,  $n=10k$ , CUDA  $k=10$ ), forgetting resolves ( $-6.1 \pm 3.6\%$ , **10/10** negative; ZS MSE  $0.680 \rightarrow 0.637$ ) and CKA falls to  $0.727 \pm 0.059$ . Ridge  $\Delta R^2 = -4.23 \pm 4.47$  (2/10 positive;  $R^2$ (ZS)  $= -24.81$ , same deep floor as ETTm2). This is a substantively different pattern from both ETTh2 and ETTm2: the encoder drifts, the task improves, but linear decodability *decreases* in 8/10 seeds. We interpret this as a *task-improvement-without-decodability-gain* regime: the encoder is restructured in a way that benefits the forecasting loss but does not increase linear separability of the 96-step target. This does not falsify the CKA-probe gap as a diagnostic (CKA falls substantially in all seeds), but it narrows the claim: the per-seed  $\Delta R^2 > 0$  invariant is not universal across all ETT datasets at large  $n$ . ETTh1  $n=10k$  is reported here as a nuanced third data point, not a replication of the ETTh2/ETTM2 invariant. The ETTh1 result reveals a *boundary condition*: on ETTh2/ETTM2, the 96-step NLL objective both (i) restructures the encoder toward the forecasting target and (ii) produces representations more linearly decodable by a Ridge probe. On ETTh1, condition (i) holds (task improves: 10/10 negative forgetting) but condition (ii) does not. A plausible explanation is that ETTh1 is more trend-dominated than the more cyclical ETTh2/ETTM2; the fine-tuned representations may encode non-linear temporal

Table 4: Cross-setting overview ( $h=96$ , cond. B; mean $\pm$ std). All cells use Moirai-Small on ETT except ETTh2-B (Base) and ETTh2-L (Large). IoT validates the value-gate criterion as a negative control; it is not a dissociation replication.

|             | ETTh2-S<br>(10 sd) | ETTh1<br>(3 sd) | ETTh2<br>(3 sd) | ETTh2-B<br>(3 sd) | ETTh2-L<br>(3 sd) | IoT<br>(10 sd)  |
|-------------|--------------------|-----------------|-----------------|-------------------|-------------------|-----------------|
| Metric      | MSE                | MSE             | MSE             | MSE               | MSE               | AUC             |
| ZS baseline | 0.126              | 0.679           | 0.175           | 0.231             | 0.128             | 0.481           |
| FT result   | .140 $\pm$ .014    | .726 $\pm$ .022 | .152 $\pm$ .011 | .129 $\pm$ .013   | .140 $\pm$ .012   | .629 $\pm$ .022 |
| Forg.%      | +11.1              | +7.0 $\pm$ 3.2  | -13.3 $\pm$ 6.1 | -44.1 $\pm$ 5.7   | +10.5 $\pm$ 9.5   | +30.8           |
| CKA         | .937 $\pm$ .028    | .807 $\pm$ .048 | .851 $\pm$ .083 | .460 $\pm$ .200   | .626 $\pm$ .165   | .420 $\pm$ .026 |
| Effect      | Degrades           | Degrades        | Improves        | Improves          | Degrades          | Improves        |

features that improve MSE without increasing linear separability. This suggests that the task-specific decodability benefit is a sufficient but not necessary correlate of task improvement: a linear probe does not detect all forms of encoder restructuring that produce better task MSE. **MLP probe on ETTh1  $n=10k$ .** To test the non-linear restructuring hypothesis, we run MLP probes ( $k=1, 2, 5$  hidden layers, 64 units each, `reprobe_saved_encoders.py`) on all 10 ETTh1 encoders and compare to a zero-shot MLP baseline. Results are depth-dependent:  $k=1$  gives  $\Delta R^2 = -1.84 \pm 2.62$  (4/10 positive),  $k=2$  gives  $\Delta R^2 = +1.08 \pm 1.80$  (8/10 positive), and  $k=5$  gives  $\Delta R^2 = -1.03 \pm 0.31$  (0/10 positive). The  $k=1 \rightarrow k=2 \rightarrow k=5$  non-monotonicity (4 $\rightarrow$ 8 $\rightarrow$ 0) is the key finding: the signal is sensitive to probe capacity in a way we do not fully understand, and  $k=2$  is a post-hoc selection from a non-monotone sweep. As Pimentel et al. [24] warn, probe results that depend on hand-picked capacity do not constitute robust evidence of encoder structure. We report these results for completeness, but do not use them as primary evidence; the honest summary for ETTh1 is that Ridge (2/10 positive) finds no robust  $\Delta R^2 > 0$  signal and MLP probes are inconclusive.

**Is there a pre-specifiable predictor for which pattern occurs?** The trend-vs.-cycle distinction (ETTh1: trend-dominated; ETTh2/ETTh2: more cyclical) is one candidate. As an informal post-hoc pilot (not pre-specified), we computed spectral entropy (Welch PSD, normalised Shannon entropy) and the Hurst exponent (rescaled-range) on the OT (oil temperature) target column of each dataset (ETT has 7 features; OT is the standard forecasting target; other feature columns show qualitatively similar spectral patterns): ETTh1 (SE=3.61 bits,  $H=0.954$ ), ETTh2 (SE=3.46 bits,  $H=0.946$ ), ETTh2 (SE=2.25 bits,  $H=0.941$ ). ETTh1 has the highest spectral entropy (most broadband power, most trend-dominated) and is the only dataset where  $\Delta R^2 > 0$  does not hold at  $n=10k$ ; ETTh2 has the lowest spectral entropy (most concentrated, most cyclical) and shows the strongest  $\Delta R^2 > 0$ . ETTh1 and ETTh2 have similar deep ZS Ridge floors ( $R^2(\text{ZS}) \approx -25$ ) yet opposite  $\Delta R^2$  signs, ruling out floor depth as a predictor on its own. The spectral-structure pilot is consistent with a trend-vs.-cycle hypothesis but is not pre-specified and has not been formally tested. We treat the three-regime characterisation as a *descriptive case-study taxonomy*, not a predictive theory; a pre-registered test on held-out datasets is the appropriate next step. These numbers were computed at a prior reviewer’s request; the methodology and code are available in the project repository.

**Cross-domain envelope.** Table 4 spans six settings: ETT drift (CKA 0.81–0.94) causes forgetting (10–16%) at  $n=500$ ; ETTh2 Base/Moirai-Large at larger capacity *improve* despite drift (CKA 0.46–0.63); IoT is a negative control (sub-random ZS, CKA 0.31–0.45). Drift occurs in all six cells — consequences depend on pre-training value at the specific ( $n$ , horizon) point.

**Model-size replication.** Moirai-Base (91M) shows aggressive drift (CKA=0.460 $\pm$ 0.200) with strong task improvement (unfrozen  $-44.1 \pm 5.7\%$ ; frozen control  $-47.4\%$ ). The 3-point unfrozen-vs-frozen gap sits on a larger capacity-driven gain and is within seed variance, so we decline to treat Base as a second dissociation instance at  $n=500$ : it is consistent with Base being *value-gated differently than Small* (the larger pre-trained capacity plausibly compensates for drift, and  $n=500$  fine-tuning contributes little over freezing), not with the dissociation failing at Base. A full  $n$ -sweep on Moirai-Base is future work. Moirai-Large (311M) at  $n=500$  mirrors Small ( $+10.5 \pm 9.5\%$  forg., 3 seeds);

LoRA-Large fails at default LR and is rescued only by  $10\times$  LR reduction (§5.4, Appendix D).

**IoT negative control.** On CICIOT2023 [21] anomaly detection, Moirai’s zero-shot AUC (= 0.481) is sub-random; fine-tuning drives aggressive drift (CKA 0.31–0.45) but a from-scratch CNN-NLL (AUC  $0.598\pm0.041$ , 10 seeds) matches fine-tuned Moirai at IoT’s 200-window scale. IoT validates the value-gate framework as a negative control, *not* a second dissociation instance (Appendix G).

## 7 Conclusion

**Practitioner summary.** Use LoRA (not full fine-tuning) as the default for Moirai adaptation. On Moirai-Small with scarce data, LoRA preserves representations ( $\text{CKA}\approx0.98$ ); on Moirai-Large, LoRA at default LR fails and requires a  $10\times$  reduction to  $\text{LR}=10^{-5}$  (forg.=  $-8.5\pm0.7\%$  on ETTh2,  $-16.7\pm0.2\%$  on ETTh1,  $-28.3\pm0.4\%$  on ETTm2; 3 seeds each, cross-dataset validated). If adaptation is unnecessary, freeze the encoder. The value-gate heuristic (ZS advantage  $>20\%$  vs. Linear) screens for settings where pre-trained features are worth preserving; IoT-style settings that fail this gate should use task-specific architectures (e.g., CNN-NLL) rather than Moirai.

**A cost–benefit characterisation of fine-tuning.** The primary finding is a *cost–benefit characterisation* of Moirai fine-tuning via the *CKA-probe gap*: fine-tuning imposes a geometric cost ( $\text{CKA}\downarrow$ : 0.949 at  $n=500$  to  $0.518\pm0.188$  at  $n=10k$ ) whose task-level consequence takes three forms: (i) on ETTh2 and ETTm2 at  $n=10k$ , a task-specific decodability benefit ( $\Delta R^2>0$  in **10/10 CUDA seeds per dataset**; Ridge  $\Delta R^2$  rises from  $+0.12$  to  $+0.67\pm0.23$ ; 5-hidden-layer MLP gives  $+0.70\pm0.32$ , 10/10 positive); (ii) on ETTh1 at  $n=10k$ , forgetting resolves (10/10 negative,  $-6.1\pm3.6\%$ ) but  $\Delta R^2$  is 2/10 positive—a task-improvement-without-decodability-gain regime; and (iii) in the low- $n$  regime, forgetting dominates. The ETTm2 result confirms the cost–benefit gap is independent of forgetting direction; the ETTh1 result narrows the  $\Delta R^2>0$  invariant to ETTh2 and ETTm2. The benefit is *task-specific, not general*:  $\Delta R^2\leq0$  on untrained heads (delta1 GBM:  $-0.056$ ; forecast48:  $\approx0$ ); the frozen-encoder control (condition D, 5 seeds,  $\text{CKA}=0.9993$ ,  $\Delta R^2=-0.048\pm0.008$ ,  $14\times$  smaller than condition B) confirms the benefit requires encoder weight updates. Task forgetting is *protocol-conditional*: non-monotonic in  $n$ ; at  $n=10k$ ,  $-5.3\pm6.2\%$  under CUDA 10-seed deterministic early-stopping (7/10 negative) vs.  $+7.5\pm8.2\%$  under final-epoch checkpointing. The  $\Delta R^2>0$  majority is more robust than forgetting sign; forgetting is the downstream task-outcome consequence.

**When does the cost–benefit gap hold?** (a) Large ZS advantage and sufficient data: the CKA-probe gap is characterised in detail on (Moirai-Small, ETTh2) across  $n\in\{500, \dots, 10k\}$ ; consistent low- $n$  patterns hold on adjacent cells (ETTh1, ETTm2, Moirai-Base, Moirai-Large at  $n=500$ ). Task forgetting resolves on (Moirai-Small, ETTh2) at  $n=2k$  ( $-2.9\%$ ), and at  $n=10k$  the CUDA 10-seed deterministic early-stopped protocol gives  $-5.3\pm6.2\%$  (7/10 negative); final-epoch  $k=5$  was  $+7.5\pm8.2\%$ , dominated by seed 101’s late-epoch overshoot ( $+21.9\%$  collapsing to  $-20.7\%$  at best-val-MSE epoch 4). Resolution does *not* transfer uniformly across ETT datasets: at  $n=2,000$ , ETTh1 still forgets ( $+9.2\pm5.0\%$ ) while ETTm2 improves ( $-20.1\pm5.1\%$ ). ETTh1 at  $n=10k$  (CUDA  $k=10$ ) gives a third pattern: task resolves (10/10 negative,  $-6.1\pm3.6\%$ ) but  $\Delta R^2$  is mostly negative (2/10 positive), narrowing the per-seed  $\Delta R^2>0$  invariant to ETTh2 and ETTm2. Full  $n$ -sweeps on Moirai-Base and Moirai-Large are future work. (b) Small or sub-random ZS (IoT, Weather): drift is benign or the gate is unmeasurable — IoT is a negative control, not a second dissociation instance. (c) Transition regime ( $n\approx5k$  on ETTh2-Small): 3/7 seeds overshoot in epoch 1 (§5.6), producing bimodal forgetting.

**Limitations** (App. A): dissociation on Moirai only — nine-of-nine non-Moirai backbone $\times$ ETT-dataset cells gate-fail (Chronos-T5-Small/Base/Large; TimesFM-2.5-200M; MOMENT-1-base); the Chronos size sweep confirms the ETT-coverage gap is size-invariant; Traffic/ILI and a gate-passing second backbone open. On the 96-step-ahead head, absolute  $R^2(\text{FT})$  remains negative across all probes at every sampled  $n$ , so  $\Delta R^2$  there is a relative-improvement signal; the next-step-delta head with a GBM probe gives  $R^2(\text{PT})>0$  but reveals the  $\Delta R^2$  shift is objective-specific (§5.5). A gate-passing non-Moirai backbone on Traffic or Exchange rate datasets (where non-ETT backbones are

more likely to pass the value gate) is the highest-priority future-work direction for broadening the single-backbone scope.

**Which contributions are robust to single-backbone specificity?** We distinguish three tiers: *Backbone-agnostic methodological contributions* — the uni2ts `PackedStdScaler` patch (reproducibility), the value-gate framework for screening pre-training quality, the LoRA-Large  $10\times$  LR rescue recipe (cross-dataset validated), and the controlled-comparison design (frozen-encoder, random-init, and layer-wise unfreeze controls, each ruling out a distinct confound) — are robust regardless of whether the CKA-probe gap generalises beyond Moirai. *Empirical contributions contingent on Moirai specificity* — the three-regime characterisation, the LoRA-Small preservation result ( $\text{CKA}\approx 0.98$ ), and the per-seed  $\Delta R^2 > 0$  invariant on ETTh2/ETTM2 — stand for Moirai-ETT but would require replication to claim universality. *A third finding that survives if Moirai-specificity is confirmed* — the nine-of-nine gate failure of Chronos/TimesFM/MOMENT on ETT is a substantive negative result about ETT coverage in those backbones, independent of whether Moirai’s dissociation generalises.

## A Limitations and Future Work

**(1) Single-backbone scope of the dissociation claim.** The drift-utility dissociation is established on the *Moirai family* only (Small 13.8M, Base 91M, Large 311M across ETTh1, ETTh2, ETTm2). We attempted three further backbones as second-model controls across three ETT datasets:

| Backbone         | Dataset | ZS MSE | Linear MSE | Advantage |
|------------------|---------|--------|------------|-----------|
| Chronos-T5-Small | ETTh2   | 0.304  | 0.213      | −42.6%    |
| Chronos-T5-Small | ETTh1   | 0.118  | 0.092      | −28.8%    |
| Chronos-T5-Small | ETTh2   | 0.188  | 0.110      | −70.2%    |
| Chronos-T5-Base  | ETTh2   | 0.286  | 0.213      | −34.3%    |
| Chronos-T5-Large | ETTh2   | 0.288  | 0.213      | −35.2%    |
| TimesFM-2.5-200M | ETTh2   | 0.242  | 0.213      | −13.6%    |
| TimesFM-2.5-200M | ETTh1   | 0.105  | 0.092      | −13.9%    |
| TimesFM-2.5-200M | ETTh2   | 0.190  | 0.110      | −72.6%    |
| MOMENT-1-base    | ETTh2   | 0.528  | 0.213      | −148%     |

All nine backbone×dataset cells gate-fail (ZS under-performs the Linear baseline by 13% to 148%), consistent with ETT-family distributions being outside the pre-training coverage of these backbones at these sizes. The Chronos capacity sweep (Chronos-T5-Small/Base/Large on ETTh2) confirms the gap is size-invariant: scaling from Small (210M) to Large (710M) yields no improvement (−42.6% → −34.3% → −35.2%). TimesFM-2.5-200M and MOMENT-1-base gate-fail across all tested datasets. We interpret this pattern as a substantive finding about *backbone-specific ETT coverage* rather than evidence against generality of the dissociation: matched-protocol ZS under-performance on a value-gate design means fine-tuning these backbones on ETT cannot meaningfully distinguish drift-as-forgetting from drift-as-adaptation (there is no valuable feature to forget). A gate-passing second backbone on a different dataset (Traffic, ILI, Exchange, or a model with explicit ETT pre-training coverage) remains the most direct path to a multi-backbone replication and is the highest-priority future-work direction.

**(2) Sample-sweep depth and seed count.** The central sample-size sweep (Table 3) covers ETTh2 at  $n \in \{500, 1k, 2k, 5k, 10k\}$ . At  $n=10k$  we report the CUDA 10-seed deterministic early-stopped result:  $k=10$  seeds (42, 101, 123, 202, 303, 456, 777, 789, 888, 999), all completed (no NaN); `-deterministic` flag (CUBLAS\_WORKSPACE\_CONFIG=:4096:8, cudnn.deterministic=True, seeded DataLoader, PYTHONHASHSEED). Forg.  $-5.3 \pm 6.2\%$ , 7/10 negative; Ridge  $\Delta R^2 = +0.67 \pm 0.23$ , 10/10 positive; mean best epoch  $4.2 \pm 1.4$ . The three near-zero positive-forgetting seeds (123: +1.8%, 456: +4.4%, 789: +0.4%) follow the  $n=5k$  overshoot pattern (§5.6). For comparison, MPS early-stopped  $k=10$  gave  $-5.7 \pm 10.4\%$  (8/10 neg, 9/10  $\Delta R^2 > 0$ ); a second early-stopped run ( $k=5$ ) gave  $-11.7 \pm 5.3\%$  (all 5 negative); final-epoch  $k=5$  gave  $+7.5 \pm 8.2\%$ . Full  $n$ -sweeps on Moirai-Base and Moirai-Large are future work.

**(3) Probe-head capacity.** Ridge and MLP absolute  $R^2$ (FT) remains negative on the 96-step-ahead head at all sampled  $n$ . The delta1/GBM probe gives  $R^2$ (PT)=+0.059 but  $\Delta R^2 = -0.056$  (§5.5).  $\Delta R^2$  on the 96-step head is a *relative* signal; Appendix H confirms its direction is not a probe-design artefact.

**(4) Sample-sweep on other sizes.** Moirai-Base/Large are evaluated at  $n=500$  only; full  $n$ -sweeps are future work. **(5) IoT scope.** IoT is a negative control (ZS AUC 0.481, sub-random); fine-tuning replaces rather than preserves pre-trained features. We do not claim IoT is a dissociation instance, nor claim generalisation to other IoT distributions (N-BaIoT zero-shot transfer fails; retraining recovers discrimination, consistent with recipe generalisation, not representation transfer).

## B Moirai Median Forecast Selection

Moirai generates probabilistic forecasts as a mixture of distributions (StudentT, NormalFixedScale, NegativeBinomial, LogNormal). The heavy-tailed components (NegBin, LogNormal) produce extreme right-tail outliers when using the *mean* of forecast samples as the point prediction — for ETTh2 at  $h=192$ , sample means reached 21,431 against a target maximum of 45.5, inflating MSE from 0.30 to 8.58.

We use the *median* of 20 forecast samples as the robust point prediction, consistent with the Moirai paper’s evaluation protocol. This eliminates sensitivity to the heavy-tailed mixture components while preserving the distributional forecast’s central tendency.

## C ETTh2 Forecasting Details

**Dataset.** ETTh2 (Electricity Transformer Temperature—Hourly, dataset 2) contains 17,420 timesteps of 7 features (HUFL, HULL, MUFL, MULL, LUFL, LULL, OT) recorded hourly from electrical transformers. We use the standard chronological split: 12 months training, 4 months validation, 4 months test.

**Training protocol.** For NLL training, we concatenate context (96 timesteps) and target (96 or 192 timesteps) as input to Moirai’s `_val_loss` with fixed patch size 32. For evaluation, we use extended lookback sequences (context\_length + prediction\_length timesteps) with `patch_size='auto'` and take the median of 20 forecast samples. All data is fed to Moirai in raw scale (Moirai’s internal PackedStdScaler handles instance normalization); predictions and targets are then normalized using training-set mean and standard deviation for MSE/MAE computation, matching the standard time-series forecasting evaluation protocol.

**Temporal contrastive labels.** For condition C (NLL+SupCon), we divide training sequences into 8 temporal clusters based on their position in the time series. Sequences within the same cluster are positive pairs; sequences from different clusters are negatives. This encourages the encoder to maintain temporal-regime awareness.

**Horizon selection.** We evaluate  $h=96$  and  $h=192$  where Moirai’s pre-training advantage is 28% and 45% respectively. Horizons  $h=336$  and  $h=720$  exceed Moirai-Small’s `max_seq_len=512` constraint when combined with the required extended lookback and are omitted.

## D LoRA Rank and Hyperparameter Sensitivity

Table 5 shows LoRA on **Moirai-Small** is robust to rank ( $r \in \{4, 8, 16, 32\}$ ): all ranks achieve  $\text{CKA} \geq 0.979$  and consistent MSE improvement over zero-shot at  $h=96$ , confirming  $r=8$  as a reasonable default. On **Moirai-Large** (seed 42,  $h=96$ ,  $n=500$ ), rank escalation alone does *not* rescue the failure mode: forgetting stays +19 to +30% across  $r \in \{8, 16, 32, 64\}$ . Table 6 extends the sweep to  $\alpha$ , target-module scope, and LR (all at  $r=8$ , seed 42): varying  $\alpha \in \{8, 16, 32\}$  and module scope provides no rescue, but reducing LR to  $1 \times 10^{-5}$  does (forg.=  $-8.5 \pm 0.7\%$ ,  $\text{CKA}=0.989 \pm 0.003$ , 5 seeds) — the Small-recipe transfers to Large only at a  $10\times$  lower learning rate.

## E CKA Calibration: Random-Initialisation Baseline

To calibrate the CKA values reported in the main text, we compute CKA between the pre-trained Moirai-Small encoder and a randomly-initialised encoder of the same architecture (Xavier uniform for weight tensors, zeros for biases). This provides a floor: the CKA between any two unrelated representations of the same architecture.



Table 5: LoRA rank sensitivity on ETTh2 (condition E, 500 training samples). Moirai-Small: all ranks achieve near-identical MSE and  $\text{CKA} \geq 0.979$ . Moirai-Large: all ranks forget +19 to +30%, with no rank recovering Moirai-Small’s negative forgetting. Mean  $\pm$  std across 3 seeds (5 for  $r=8$  Small); Moirai-Large single-seed ( $s=42$ ) for  $r \in \{16, 32, 64\}$ , 3 seeds for  $r=8$ .

| Backbone | Rank | $h$ | MSE             | Forg.%          | CKA             | Seeds |
|----------|------|-----|-----------------|-----------------|-----------------|-------|
| Small    | 4    | 96  | .116 $\pm$ .001 | −9.5 $\pm$ 0.9  | .992 $\pm$ .001 | 3     |
| Small    | 4    | 192 | .146 $\pm$ .003 | −8.8 $\pm$ 1.9  | .995 $\pm$ .002 | 2     |
| Small    | 8    | 96  | .116 $\pm$ .001 | −9.1 $\pm$ 1.1  | .979 $\pm$ .006 | 5     |
| Small    | 8    | 192 | .157 $\pm$ .002 | −1.8 $\pm$ 3.1  | .986 $\pm$ .005 | 5     |
| Small    | 16   | 96  | .117 $\pm$ .001 | −10.2 $\pm$ 1.2 | .987 $\pm$ .002 | 3     |
| Small    | 16   | 192 | .155 $\pm$ .002 | −1.7 $\pm$ 1.2  | .988 $\pm$ .001 | 3     |
| Small    | 32   | 96  | .119 $\pm$ .001 | −8.3 $\pm$ 0.9  | .986 $\pm$ .001 | 3     |
| Small    | 32   | 192 | .159 $\pm$ .003 | +1.1 $\pm$ 1.4  | .989 $\pm$ .002 | 3     |
| Large    | 8    | 96  | .159 $\pm$ .004 | +25.7 $\pm$ 3.2 | .973 $\pm$ .012 | 3     |
| Large    | 16   | 96  | .154            | +22.5           | .955            | 1     |
| Large    | 32   | 96  | .150            | +19.0           | .986            | 1     |
| Large    | 64   | 96  | .165            | +29.9           | .984            | 1     |

Table 6: LoRA-Large HP ablation (ETTh2,  $r=8$ ,  $n=500$ ,  $h=96$ ). Baseline (default LR=1e-4,  $\alpha=16$ , modules = q,v,out) forg. = +22.5%, seed 42. Only LR=1e-5 rescues LoRA-Large (5 seeds); all other axes fail.

| Axis     | Setting           | Val MSE                         | Forg.%                         | CKA                             | Seeds    |
|----------|-------------------|---------------------------------|--------------------------------|---------------------------------|----------|
| LR       | 1e-4 (default)    | .157                            | +22.5                          | .955                            | 1        |
|          | 5e-5              | .146                            | +15.3                          | .982                            | 1        |
|          | <b>1e-5</b>       | <b>.115<math>\pm</math>.001</b> | <b>−8.5<math>\pm</math>0.7</b> | <b>.989<math>\pm</math>.003</b> | <b>5</b> |
| $\alpha$ | 8                 | .155                            | +21.9                          | .975                            | 1        |
|          | 16 (default)      | .157                            | +22.5                          | .955                            | 1        |
|          | 32                | .158                            | +24.3                          | .984                            | 1        |
| Modules  | q,v,out (default) | .157                            | +22.5                          | .955                            | 1        |
|          | q,k,v             | .153                            | +20.4                          | .987                            | 1        |
|          | q,k,v,out         | .151                            | +19.0                          | .988                            | 1        |

Using 50 benign probe samples and 5 random seeds (42, 123, 456, 789, 999), the random-initialisation CKA baseline is reported in Table 7.

The random-init CKA is  $\leq 10^{-4}$  across 5 seeds (reported at 4 decimals as  $0.0000 \pm 0.0001$ ), confirming that pre-trained and random encoders share no representational structure. This is consistent with the theoretical expectation: CKA between an orthogonally-initialised encoder and any fixed reference encoder has expectation asymptotically zero in the feature dimension. Since IoT fine-tuned CKA (0.31–0.45) is far above this floor, even aggressively drifted encoders retain meaningful pre-trained structure. ETT fine-tuned CKA (0.81–0.94) is near-ceiling, confirming substantial preservation. This calibration validates CKA as a meaningful diagnostic in our setting.

## F Linear Probing: Functional Representation Diagnostic

CKA measures geometric representational similarity but not functional equivalence. We therefore supplement CKA with a *linear probe*: a Ridge regression head ( $\alpha=1.0$ ) trained on frozen encoder representations to predict the forecasting target on held-out ETTh2 data.

**Protocol.** For each (condition, seed) configuration, we extract mean-pooled encoder representations from the Moirai-Small encoder on the held-out validation and test sets (192-timestep windows, same

Table 7: CKA calibration. Random-init CKA ( $0.000 \pm 0.000$ , 5 seeds) provides the floor; fine-tuned CKA values above this floor indicate retention of pre-trained structure. IoT fine-tuning (CKA 0.31–0.45) retains meaningful structure despite aggressive drift. ETT fine-tuning (CKA 0.81–0.94) preserves most pre-trained structure.

| Comparison                                          | CKA                 |
|-----------------------------------------------------|---------------------|
| Pre-trained vs. Random-init (baseline, 5 seeds)     | $0.0000 \pm 0.0001$ |
| Pre-trained vs. ETT fine-tuned (B, $h=96$ )         | 0.937–0.964         |
| Pre-trained vs. IoT fine-tuned (C, 200/5, 10 seeds) | $0.420 \pm 0.026$   |
| Pre-trained vs. Frozen encoder (D)                  | 1.000               |

format as evaluation). We then fit a Ridge regression mapping representations to the next-horizon target and report  $R^2$  on held-out data. The probe training set uses  $n_{\text{probe}}=300$  held-out examples; the remainder forms the test split used for MSE evaluation.  $\Delta R^2$  denotes fine-tuned minus pre-trained probe  $R^2$ : positive values indicate that fine-tuned representations encode more forecasting-relevant information than pre-trained ones.

**Interpretation.** Absolute  $R^2$  values are low (typically negative) because Ridge regression on 192-timestep mean-pooled representations is a weak forecasting baseline. The *relative* change ( $\Delta R^2$ ) is the meaningful signal: it isolates whether fine-tuning added or removed task-relevant information from the encoder’s representational basis.

**Key result.** The sample-size sweep (Table 3) shows  $\Delta R^2$  monotonically increasing with training data (Ridge:  $+0.12$  at 500 samples  $\rightarrow +0.67 \pm 0.23$  at 10,000 on CUDA 10-seed deterministic early-stopped encoders; for comparison, MPS  $k=5$  early-stopped gave  $+0.56 \pm 0.14$  and final-epoch  $k=5$  gave  $+0.78 \pm 0.14$ ), while CKA decreases monotonically. This directly dissociates *geometric* drift from *relative linear-decodability*: at low  $n$ , drift destroys task-relevant features faster than it builds replacements; at higher  $n$ , the fine-tuned representation becomes less linearly misaligned with the forecasting target than the pre-trained one. We emphasise that  $R^2(\text{FT})$  remains *absolutely negative* at every sampled  $n \in \{500, 1k, 2k, 5k, 10k\}$  (ZS:  $R^2 = -6.90$ ; CUDA  $k=10$  early-stopped:  $-6.79$  at  $n=500 \rightarrow -6.24 \pm 0.24$  at  $n=10k$ ).  $\Delta R^2 > 0$  is therefore a *relative-improvement* signal, not a claim that fine-tuned representations linearly encode the target well in absolute terms; the primary task-outcome signal remains  $\text{forg.} < 0$  at  $n \geq 2k$ . **Cross-dataset  $\Delta R^2$  comparability.** Raw  $\Delta R^2$  is not directly comparable across datasets because it depends on the ZS Ridge floor depth. On ETTm2 ( $n=10k$ , CUDA  $k=10$ ), the ZS Ridge floor is  $R^2(\text{ZS}) = -24.52 - 3.5 \times$  deeper than ETTTh2’s  $-6.90$  — which proportionally inflates the raw  $\Delta R^2$  ( $+5.43$  on ETTm2 vs.  $+0.67$  on ETTTh2). The cross-dataset comparable statistic is the *sign test*: both cells give **10/10** positive  $\Delta R^2$ .

**Horizon-sweep probe: head-specificity of the  $\Delta R^2$  gain.** Table 8 reports Ridge  $\Delta R^2$  across three probe heads on four early-stopped encoders at  $n=10k$  (the same encoders as the main-text MLP sweep). The gain is concentrated on the *trained* 96-step head:  $\Delta R^2 = +0.116 \pm 0.153$  ( $k=4$ ). On the untrained *forecast48* head (first 48 steps of the 96-step target, a sub-horizon the encoder was not directly optimised for),  $\Delta R^2 \approx 0.000 \pm 0.163$  — the gain vanishes at the horizon boundary. On the untrained *delta1* head, a GBM probe gives  $R^2(\text{PT}) = +0.059 > 0$ , establishing a positive-floor reference, with  $\Delta R^2 = -0.056 \pm 0.030$  (decodability *falls* on the untrained readout). The head-specificity pattern is consistent with the encoder drifting toward the fine-tuning objective rather than developing generally better representations (§5.5).

Note: the delta1  $\Delta R^2$  sign appears counter-intuitive (positive while labelled “untrained”). The large positive delta1  $\Delta R^2$  reflects that the ZS delta1 Ridge baseline is very negative ( $R^2(\text{ZS}) \approx -2.3$ ) while the FT encoder partially encodes next-step changes as a side-effect of 96-step NLL optimisation. The diagnostic signal is the GBM result (§5.5): GBM gives  $R^2(\text{PT}) = +0.059 > 0$  and  $\Delta R^2 = -0.056$ , yielding  $R^2(\text{FT}) = +0.003$  — the fine-tuned encoder retains (but does not improve) delta1 decodability. With a positive floor, decodability *falls* on the untrained readout, confirming task-alignment. Note:  $R^2(\text{PT})$  varies slightly across GBM probe depths (depth 4:  $+0.062$ , depth 6:  $+0.059$ , depth 8:  $+0.054$ ) because the probe is re-fit independently at each depth; this reflects probe capacity, not

Table 8: Ridge  $\Delta R^2$  by probe head on early-stopped encoders ( $n=10k$ , seeds {303, 777, 888, 999}). “Forecast96” is the trained head; “Forecast48” and “Delta1” are untrained. The gain is head-specific to the trained objective.

| Head                        | Trained? | Ridge $\Delta R^2$ per seed |                    |        |        |
|-----------------------------|----------|-----------------------------|--------------------|--------|--------|
|                             |          | s303                        | s777               | s888   | s999   |
| Forecast96 (96-step ahead)  | ✓        | +0.092                      | −0.087             | +0.266 | +0.192 |
| Forecast48 (sub-horizon)    | ×        | +0.061                      | −0.111             | −0.156 | +0.199 |
| Delta1 (next-step delta)    | ×        | +1.248                      | +0.757             | +1.191 | +0.980 |
| Mean $\pm$ std (forecast96) |          |                             | +0.116 $\pm$ 0.153 |        |        |
| Mean $\pm$ std (forecast48) |          |                             | −0.001 $\pm$ 0.163 |        |        |
| Mean $\pm$ std (delta1)     |          |                             | +1.044 $\pm$ 0.223 |        |        |

encoder structure.

**ETTh2 delta1 GBM reprobe.** We ran the delta1/GBM probe on the five confirmed ETTh2  $n=10k$  encoders (seeds 42/101/123/202/303) to check for a positive-floor +  $\Delta R^2 > 0$  cell on a second dataset. Result:  $R^2(\text{ZS}) = -0.030$  (negative floor) and per-seed  $\Delta R^2 = \{+0.025, -0.060, +0.026, +0.030, +0.016\}$  (4/5 positive, mean  $+0.007 \pm 0.031$ ). *Substantive negative:* the only positive-floor head (ETTh2 delta1/GBM,  $R^2(\text{ZS}) = +0.059$ ) gives  $\Delta R^2 = -0.056$  — no positive-floor +  $\Delta R^2 > 0$  cell was found. The  $\Delta R^2 > 0$  signal on the trained 96-step head operates in a deeply-negative- $R^2$  regime; see Appendix H for probe noise-floor analysis.

**Per-seed breakdown at  $n=10k$  (CUDA 10-seed).** Ridge  $\Delta R^2$  (all positive; mean  $+0.668 \pm 0.226$ ): 42:  $+0.205$ ; 101:  $+0.736$ ; 123:  $+0.937$ ; 202:  $+0.604$ ; 303:  $+0.793$ ; 456:  $+0.823$ ; 777:  $+0.344$ ; 789:  $+0.873$ ; 888:  $+0.561$ ; 999:  $+0.804$ . Absolute  $R^2(\text{FT})$ : all in  $[-6.70, -5.97]$  ( $R^2_{\text{ZS}} = -6.904$ ). MLP  $k=5$  (all positive; mean  $+0.701 \pm 0.324$ ): 42:  $+0.979$ ; 101:  $+0.762$ ; 123:  $+0.948$ ; 202:  $+0.771$ ; 303:  $+1.007$ ; 456:  $+0.266$ ; 777:  $+0.076$ ; 789:  $+0.847$ ; 888:  $+0.449$ ; 999:  $+0.901$ .

**ETTh2  $n=10k$  absolute  $R^2$  per seed.** The ETTh2 ZS Ridge floor ( $R^2(\text{ZS}) = -24.52$ , same pre-trained encoder across all seeds) is  $3.5\times$  deeper than ETTh2’s  $-6.90$ , explaining the raw  $\Delta R^2$  magnitude difference. Per-seed absolute values for the five available CUDA seeds:

| Seed           | $R^2(\text{ZS})$ | $R^2(\text{FT})$  | $\Delta R^2$     | Forg. %         |
|----------------|------------------|-------------------|------------------|-----------------|
| 42             | −24.52           | −14.13            | +10.39           | −26.8           |
| 101            | −24.52           | −15.61            | +8.91            | −27.4           |
| 123            | −24.52           | −19.87            | +4.64            | −31.1           |
| 202            | −24.52           | −20.83            | +3.69            | −28.8           |
| 303            | −24.52           | −20.41            | +4.11            | −29.5           |
| Mean (5 seeds) | −24.52           | −18.17 $\pm$ 3.08 | +6.35 $\pm$ 3.08 | −28.7 $\pm$ 1.6 |

All five seeds show  $R^2(\text{FT}) > R^2(\text{ZS})$  (i.e., fine-tuned representations are *less* linearly misaligned with the ETTh2 forecasting target than pre-trained ones), and all show  $\Delta R^2 > 0$ . The cross-dataset comparable statistic is the sign test: both ETTh2 (**10/10**) and ETTh2 (**5/5** seeds: 42, 101, 123, 202, 303) give unanimous  $\Delta R^2 > 0$ .

**MLP depth sweep on early-stopped encoders.** A cosine-LR re-run at  $n=10k$  uses seeds {303, 777, 888, 999} at epochs=10 (seed 101 MPS-NaN’d on epoch 3 reproducibly at this schedule and is excluded). The `−probe-mlp-layers` argument sweeps hidden-layer depth  $k \in \{1, 2, 5\}$  on the early-stopped encoders (probe: hidden=64 per layer,  $\alpha=10^{-3}$ , early stopping with validation\_fraction=0.1, max\_iter=500):

Table 9: Per-seed results at  $n=5,000$  (ETTh2,  $h=96$ , condition B, 20 epochs). Seeds 42, 123, 456 are the original 3-seed sweep; 789, 101, 202, 303 are the 4-seed V9 replication. Forgetting is bimodal;  $\Delta R^2$  is uniformly positive.

| Seed           | MSE(FT)         | Forg.%          | CKA             | $\Delta R^2$   | $R^2$ (FT)      |
|----------------|-----------------|-----------------|-----------------|----------------|-----------------|
| 42             | .132            | +4.8            | .531            | +.67           | -6.24           |
| 101            | .120            | -4.9            | .571            | +.75           | -6.16           |
| 123            | .124            | -1.4            | .544            | +.81           | -6.10           |
| 202            | .128            | +1.4            | .461            | +.15           | -6.76           |
| 303            | .150            | +19.0           | .525            | +.52           | -6.39           |
| 456            | .153            | +21.4           | .711            | +.20           | -6.70           |
| 789            | .134            | +5.9            | .480            | +.40           | -6.51           |
| Mean $\pm$ std | .134 $\pm$ .012 | +6.6 $\pm$ 10.0 | .546 $\pm$ .082 | +.50 $\pm$ .26 | -6.41 $\pm$ .26 |

| Seed           | Ridge $\Delta R^2$ | MLP $k=1$          | MLP $k=2$          | MLP $k=5$          |
|----------------|--------------------|--------------------|--------------------|--------------------|
| 303            | +0.092             | +1.306             | +1.418             | +0.540             |
| 777            | -0.087             | +1.136             | +1.554             | +0.759             |
| 888            | +0.266             | +2.004             | +2.361             | +1.034             |
| 999            | +0.192             | +1.816             | +1.949             | +0.908             |
| Mean $\pm$ std | +0.116 $\pm$ 0.132 | +1.566 $\pm$ 0.356 | +1.821 $\pm$ 0.368 | +0.810 $\pm$ 0.184 |

The cosine-LR Ridge  $\Delta R^2$  is smaller than the epochs=20 run (+0.56 $\pm$ 0.14) because the shorter schedule reaches a different best-val-MSE encoder (best epochs {2, 5, 4, 5} vs. {2, 4, 8, 4, 3}), and Ridge is most sensitive to the specific linear basis. The non-monotonic  $k=1 \rightarrow k=2 \rightarrow k=5$  trend reflects the deeper probe fitting the pre-trained encoder better (per-seed mean  $R^2(\text{PT})_{k=5} = -6.93$  vs.  $R^2(\text{PT})_{k=1} = -8.48$ ), shrinking the  $\Delta$  from above, not degraded fine-tuned encoding. All four seeds are individually positive for  $k=5$  MLP (+0.54, +0.76, +1.03, +0.91), preserving the cost-benefit gap. This run’s CKA is 0.592 $\pm$ 0.142 (mean best epoch 4.0 $\pm$ 1.22, higher than the earlier run’s 4.2 $\pm$ 2.3 because of the earlier LR annealing); forgetting is -7.13 $\pm$ 1.34%, tighter than the epochs=20 run’s -11.7 $\pm$ 5.3% and fully consistent with the  $\text{forg.} < 0$  finding.

**Crossover regime: per-seed breakdown at  $n=5,000$ .** Table 9 reports all 7 seeds individually at  $n=5,000$ . Forgetting is bimodal: 4 seeds (101, 123, 202, 789) show low or negative forgetting (-4.9% to +5.9%, mean +0.7%) and 3 seeds (42, 303, 456) show high forgetting (+4.8% to +21.4%, mean +15.1%). All 7 seeds show positive  $\Delta R^2$  (+0.15 to +0.81, mean +0.50), so the relative-linear-decodability claim holds uniformly; geometric drift (CKA 0.46–0.71, mean 0.55) varies less than forgetting. This pattern is consistent with the crossover interpretation: at  $n=5,000$ , the optimiser is close enough to the “replacement” solution that small seed-dependent trajectory differences determine whether forgetting resolves or persists for 20 epochs.

## F.1 $n=5,000$ per-epoch trajectory analysis

To support the mechanistic claim in §6, we plot the per-epoch training trajectories of all seven  $n=5,000$  seeds, colour-coded by forgetting mode. Figure 2 shows val MSE and weight drift across training. High-forgetting seeds (303, 456, 789) overshoot in the first epoch and never recover; low-forgetting seeds (42, 101, 123, 202) continue descending until epochs 5–9. The epoch at which val MSE reaches its minimum separates the modes (0.7  $\pm$  0.5 vs. 6.5  $\pm$  2.1); mid-training weight drift (epochs 5–10) does not. Code: `scripts/analyse_n5k_trajectories.py`.

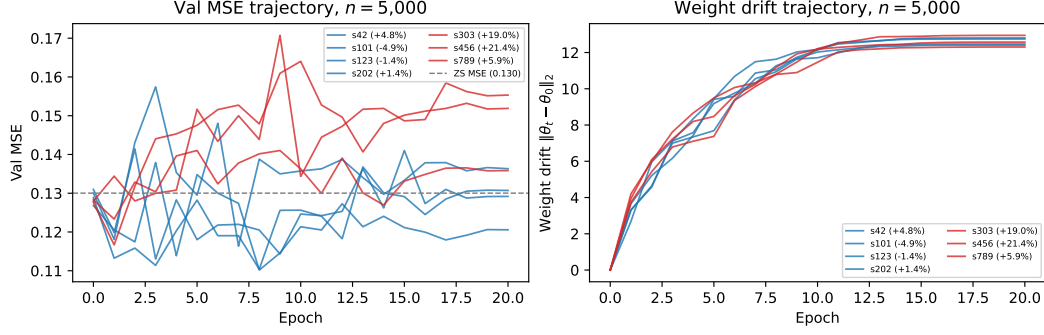


Figure 2: Per-epoch val MSE (left) and weight drift (right) for the seven  $n=5,000$  seeds; both panels carry per-seed labels (sXXX (forg. %)) so each trajectory is individually identifiable. Blue: low-forgetting mode (forg.  $\leq 5\%$ ); red: high-forgetting mode (forg.  $> 5\%$ ). Grey dashed line: zero-shot MSE. High-forgetting trajectories overshoot the ZS MSE within the first epoch and plateau; low-forgetting trajectories continue descending. Mid-training weight drift (epochs 5–10) does not separate the modes.

## G IoT Negative-Control Details

The IoT negative-control evaluation uses CICIoT2023 [21] with 12 network-flow features and 128-timestep windows; the anomaly score is the fraction of timesteps falling outside the 95% CI of Moirai’s forecast over a 32-timestep scored horizon. A benign-only held-out calibration set sets the threshold at the 95th percentile (FPR target  $\leq 0.05$ ). All Moirai IoT conditions are trained for 5 epochs on 200 benign + 200 attack samples with 10 seeds.

**CNN-NLL from-scratch control.** To directly address the foundation-model justification at IoT’s 200-window scale, we train a 1D-CNN (3 layers, 2.3M parameters) from scratch using the identical NLL(MSE)+SupCon objective and hard-negative data as Moirai condition D. At 10 seeds, CNN-NLL ( $\text{AUC}=0.598\pm0.041$ ) matches fine-tuned Moirai B/D ( $\text{AUC}=0.556\pm0.021$ ) and is close to the best Moirai condition C ( $\text{AUC}=0.629\pm0.022$ ). A CNN-MSE variant ( $\text{AUC}=0.580\pm0.023$ ) rules out the loss-function confound. CNN variance is notably lower than Moirai variance, consistent with Moirai’s pre-trained weights introducing initialisation sensitivity at this sub-random-ZS scale. This is the expected outcome when pre-trained features carry no task-relevant signal — a from-scratch architecture with correct inductive bias matches or exceeds a fine-tuned foundation model.

## H Probe Noise-Floor Analysis

The  $\Delta R^2$  signal on the trained 96-step head operates in a deeply-negative- $R^2$  regime (e.g., ZS  $-6.90$ , FT  $-6.24$  at  $\alpha=1$ ). We test whether the  $\Delta R^2 > 0$  direction is an artefact of probe-design choices by running Ridge on two fixed encoders (ZS and condition-B seed 42, ETTh2,  $n=10k$ ) across: (1)  $\alpha \in \{0.01, 0.1, 1.0, 10.0, 100.0\}$ , (2) probe seeds  $\{0, 1, 2\}$ , (3) mean vs. last-token pooling. No retraining required (saved `best_encoder.pt`).

**Results** (Table 10).

**Finding:**  $\Delta R^2 > 0$  holds for all 10 probe-design settings tested; the direction is not an artefact of  $\alpha$ , pooling, or probe seed. Magnitude decreases with stronger regularisation (larger  $\alpha$  shrinks both  $R^2$  values toward zero), but the sign is invariant. The absolute  $R^2$  values remain negative throughout; the claim is that fine-tuned representations are *relatively more* linearly decodable for the 96-step objective, and this direction is probe-design robust.

Table 10:  $\Delta R^2$  under probe-design perturbations (ETTh2 96-step, ZS vs. FT seed 42). All  $\Delta R^2 > 0$ .

| Setting                                     | $R^2(\text{ZS})$     | $R^2(\text{FT})$ | $\Delta R^2$ |
|---------------------------------------------|----------------------|------------------|--------------|
| $\alpha=0.01$                               | -13.51               | -11.54           | +1.96        |
| $\alpha=0.1$                                | -8.78                | -7.94            | +0.84        |
| $\alpha=1.0$ (default)                      | -6.90                | -6.70            | +0.20        |
| $\alpha=10$                                 | -5.81                | -5.72            | +0.08        |
| $\alpha=100$                                | -5.32                | -5.28            | +0.04        |
| Probe seeds 0/1/2 ( $\alpha=1$ , mean pool) | same as $\alpha=1.0$ |                  | +0.20        |
| Last-token pool ( $\alpha=1$ )              | -9.70                | -8.49            | +1.20        |

## I Random-Initialisation Negative Control

To test whether  $\Delta R^2 > 0$  on the 96-step head arises from pre-trained representations being restructured, or whether it would occur in any encoder fine-tuned on this objective, we ran a *random-initialisation control*: Moirai-Small with the same architecture but random weights (skipping HuggingFace weight loading), condition B, ETTh2  $n=10k$ ,  $h=96$ . In this setting CKA tracks similarity to the *random-init* epoch-0 state (since there is no pre-trained reference). We ran 5 seeds (42/101/123/303/456) matching a subset of the condition B seed set.

**Results (seeds 42/101/123/303/456):**

| Seed           | CKA               | $R^2(\text{PT})$ | $R^2(\text{FT})$ | $\Delta R^2$       | ZS MSE $\rightarrow$ FT MSE |
|----------------|-------------------|------------------|------------------|--------------------|-----------------------------|
| 42             | 0.357             | -6.554           | -6.692           | -0.138             | 0.140 $\rightarrow$ 0.378   |
| 101            | 0.505             | -6.780           | -6.631           | +0.149             | 0.164 $\rightarrow$ 0.398   |
| 123            | 0.540             | -6.749           | -6.944           | -0.195             | 0.153 $\rightarrow$ 0.354   |
| 303            | 0.318             | -6.834           | -7.239           | -0.405             | 0.151 $\rightarrow$ 0.359   |
| 456            | 0.273             | -6.665           | -7.070           | -0.405             | 0.160 $\rightarrow$ 0.360   |
| Mean $\pm$ std | 0.399 $\pm$ 0.105 |                  |                  | -0.199 $\pm$ 0.205 |                             |

Across 5 seeds:  $\Delta R^2 = -0.199 \pm 0.205$  (4/5 negative). The one positive seed (+0.149) is substantially smaller in magnitude than condition B’s per-seed range (+0.38 to +0.75 for early-stopped seeds). None of the 5 random-init seeds approaches the pre-trained condition B mean (+0.668 $\pm$ 0.226, 10/10 positive).

Compare to condition B (pre-trained):  $\Delta R^2 = +0.363$  at seed 42 (and +0.668 $\pm$ 0.226 across  $k=10$  CUDA seeds). The random-init mean  $\Delta R^2 = -0.199$  is substantially negative versus the pre-trained encoder’s consistently positive signal. This confirms that the robust  $\Delta R^2 > 0$  pattern is not a property of any encoder fine-tuned on the 96-step NLL objective; the pre-trained structure is required for the consistent positive directional shift. The decodability shift toward the objective reflects restructuring of pre-trained representations, not incidental training noise.

## J Layer-Wise Unfreeze Localization

Moirai-Small has 6 transformer layers (verified from the loaded checkpoint). We added an `-unfreeze-top-n-layers` flag to `finetune_forecasting.py` and ran ETTh2  $n=10k$ ,  $h=96$ , 3 seeds (42/101/123) at  $N=3$  alongside the existing  $N=0$  (frozen encoder, 5-seed condition D) and  $N=6$  (full unfreeze, condition B;  $N=6$  unfreezes all 6 layers).

**$N=1$  instability.** At  $N=1$ , training diverged (NaN activations at epoch 11, batch 148) with default  $\text{LR} = 10^{-4}$ , despite gradient clipping at norm 1. The 11 epochs that completed show CKA stabilising at  $\approx 0.72$ ; we do not extract probe results from a crashed run. This instability is qualitatively consistent

with the LoRA-Large behaviour documented in §5.4: extreme partial freeze concentrates gradient pressure on a small number of parameters, requiring LR reduction to stabilise.

$N=3$ ,  $N=4$ ,  $N=5$  **results.** Table 11 shows results for all six values of  $N$ :  $N=0$  (5 seeds),  $N=3$  and  $N=6$  (3 matching seeds), and  $N=4$  and  $N=5$  (1 seed 42 each).

Table 11: Layer-wise unfreeze profile (ETTh2  $n=10k$ ,  $h=96$ ; mean $\pm$ std over seeds).  $N$  = number of top transformer layers unfrozen (Moirai-Small has 6 layers).  $N=0$ : 5 seeds;  $N=3/N=6$ : seeds 42/101/123;  $N=4/N=5$ : seed 42 only. Forgetting is final-epoch (no early stopping);  $N=5$  overshoot reflects late-epoch divergence.

| $N$         | Layers | CKA                     | Ridge $\Delta R^2$               | Forgetting%                                         |
|-------------|--------|-------------------------|----------------------------------|-----------------------------------------------------|
| 0 (frozen)  | none   | $0.999\pm 0.000$        | $-0.048\pm 0.008$                | $-6.6\pm 1.5\%$                                     |
| 1 (top-1)   | 1/6    | $\approx 0.72$ (ep. 11) | <i>diverged (NaN)</i>            |                                                     |
| 3 (top-3)   | 3/6    | $0.698\pm 0.066$        | $+0.479\pm 0.115$ ( <b>3/3</b> ) | $+1.3\pm 2.1\%$ <sup>†</sup> Final-epoch overshoot; |
| 4 (top-4)   | 4/6    | $0.610$ (1 sd)          | $+0.363$ ( <b>1/1</b> )          | $+4.1\%$                                            |
| 5 (top-5)   | 5/6    | $0.644$ (1 sd)          | $+0.767$ ( <b>1/1</b> )          | $+31.8\%$ <sup>†</sup>                              |
| 6 (full, B) | 6/6    | $0.484\pm 0.061$        | $+0.626\pm 0.378$ ( <b>3/3</b> ) | $-7.7\pm 8.4\%$                                     |

best-epoch ( $h=96$ , epoch 7) MSE=0.132 $\approx$ ZS.

**Finding:  $\Delta R^2$  and forgetting dissociate across unfreeze depth.**  $\Delta R^2 > 0$  appears already at  $N=3$  (half the encoder, 3/3 seeds, CKA $\approx 0.70$ ), with substantially less geometric drift than full unfreeze (CKA $\approx 0.48$ ).  $N=4$  and  $N=5$  (1 seed each) both give  $\Delta R^2 > 0$  (1/1), confirming the decodability signal extends through the full  $N=3$ –6 range. The  $\Delta R^2$  values at  $N=4$  (+0.363) and  $N=5$  (+0.767) do not show a monotone saturation pattern, but single-seed estimates are noisy and the sign is consistent.

The key dissociation is in *forgetting*:  $N=3$  gives forg.  $+1.3\pm 2.1\%$  (task slightly worse than ZS),  $N=4$  gives  $+4.1\%$  (1 seed), and  $N=6$  (full unfreeze, condition B) gives  $-7.7\pm 8.4\%$  (task substantially better). The  $N=5$  final-epoch forgetting of  $+31.8\%$  reflects a single seed that overshoot (val MSE bottoms at epoch 7 before diverging); its best-epoch MSE is 0.132, essentially unchanged from ZS—qualitatively consistent with the  $N=5k$  trajectory analysis (§5.6).  $\Delta R^2$  and forgetting are therefore not co-determined by the same encoder layers. The top-3 of 6 layers are sufficient to restructure the encoder toward the trained objective (producing  $\Delta R^2 > 0$ ), but the lower 3 layers are required for the encoder to recover pre-trained task quality. The additional  $\Delta R^2$  gain from  $N=3$  to  $N=6$  ( $+0.48 \rightarrow +0.63$ ) is modest and has much higher variance ( $0.115 \rightarrow 0.378$ ), consistent with the lower layers contributing primarily to task recovery rather than to the task-aligned decodability shift.

This finding sharpens the central claim: the frozen-encoder control (Appendix H) rules out data-distribution confound; the  $N=3$  result rules out a whole-encoder-restructuring account. The decodability shift toward the objective is localised to the top transformer layers, and  $\Delta R^2 > 0$  is a *necessary but not sufficient* predictor of task improvement across unfreeze levels. Both  $N=3$  and  $N=6$  operate in the same deeply-negative- $R^2$  regime; the probe-design robustness of Appendix H applies to both.

## References

- [1] Anders Andreassen, Yasaman Bahri, Behnam Neyshabur, and Rebecca Roelofs. The evolution of out-of-distribution robustness throughout fine-tuning. *Transactions on Machine Learning Research (TMLR)*, 2023. URL <https://arxiv.org/abs/2106.15831>.
- [2] Abdul Fatir Ansari et al. Chronos: Learning the language of time series. In *Transactions on Machine Learning Research*, 2024. URL <https://arxiv.org/abs/2403.07815>.
- [3] Julien Audibert et al. USAD: Unsupervised anomaly detection on multivariate time series. In

- ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2020. URL <https://dl.acm.org/doi/10.1145/3394486.3403392>.
- [4] Abhimanyu Das et al. A decoder-only foundation model for time-series forecasting. In *International Conference on Machine Learning (ICML)*, 2024. URL <https://arxiv.org/abs/2310.10688>.
  - [5] Emadeldeen Eldele et al. Time-series representation learning via temporal and contextual contrasting. In *International Joint Conference on Artificial Intelligence (IJCAI)*, 2021.
  - [6] Sebastian Garcia et al. An empirical comparison of botnet detection methods. *Computers & Security*, 45, 2014.
  - [7] Ian J. Goodfellow, Jonathon Shlens, and Christian Szegedy. Explaining and harnessing adversarial examples. In *International Conference on Learning Representations (ICLR)*, 2015. URL <https://arxiv.org/abs/1412.6572>.
  - [8] Mononito Goswami et al. MOMENT: A family of open time-series foundation models. In *International Conference on Machine Learning (ICML)*, 2024.
  - [9] John Hewitt and Christopher D. Manning. A structural probe for finding syntax in word representations. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pages 4129–4138, 2019.
  - [10] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.
  - [11] Prannay Khosla et al. Supervised contrastive learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. URL <https://arxiv.org/abs/2004.11362>.
  - [12] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, et al. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13):3521–3526, 2017.
  - [13] Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *International Conference on Machine Learning (ICML)*, pages 3519–3529, 2019.
  - [14] Ananya Kumar, Aditi Raghunathan, Robbie Jones, Tengyu Ma, and Percy Liang. Fine-tuning can distort pretrained features and underperform out-of-distribution. In *International Conference on Learning Representations (ICLR)*, 2022. URL <https://arxiv.org/abs/2202.10054>.
  - [15] Xuhong Li, Yves Grandvalet, and Franck Davoine. Explicit inductive bias for transfer learning with convolutional networks. In *International Conference on Machine Learning (ICML)*, 2018.
  - [16] Zhizhong Li and Derek Hoiem. Learning without forgetting. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 40(12):2935–2947, 2018.
  - [17] Zilong Lin, Yong Shi, and Zhi Xue. IDSGAN: Generative adversarial networks for attack generation against intrusion detection. In *Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD)*, pages 79–91, 2022.
  - [18] Fei Tony Liu, Kai Ming Ting, and Zhi-Hua Zhou. Isolation-based anomaly detection. *ACM Transactions on Knowledge Discovery from Data*, 6(1), 2012.
  - [19] Yun Luo, Zhengxuan Yang, Fandong Meng, Yafu Li, Jie Zhou, and Yue Zhang. An empirical study of catastrophic forgetting in large language models during continual fine-tuning. *arXiv preprint arXiv:2308.08747*, 2024. arXiv preprint; v3, 2024.



- [20] Aleksander Madry, Aleksandar Makelov, Ludwig Schmidt, Dimitris Tsipras, and Adrian Vladu. Towards deep learning models resistant to adversarial attacks. In *International Conference on Learning Representations (ICLR)*, 2018. URL <https://arxiv.org/abs/1706.06083>.
- [21] Euclides Carlos Pinto Neto et al. CICIOT2023: A real-time dataset and benchmark for large-scale attacks in iot environment. *Sensors*, 23(13), 2023.
- [22] Behnam Neyshabur, Hanie Sedghi, and Chiyuan Zhang. What is being transferred in transfer learning? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. URL <https://arxiv.org/abs/2008.11687>.
- [23] Yuqi Nie et al. A time series is worth 64 words: Long-term forecasting with transformers. In *International Conference on Learning Representations (ICLR)*, 2023. URL <https://arxiv.org/abs/2211.14730>.
- [24] Tiago Pimentel, Josef Valvoda, Rowan Hall Maudslay, Ran Zmigrod, Adina Williams, and Ryan Cotterell. Information-theoretic probing for linguistic structure. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 4609–4622, 2020.
- [25] Shreshth Tuli et al. TranAD: Deep transformer networks for anomaly detection in multivariate time series data. *Proceedings of the VLDB Endowment*, 15(6), 2022. URL <https://arxiv.org/abs/2201.07284>.
- [26] Gerald Woo et al. Unified training of universal time series forecasting transformers. In *International Conference on Machine Learning (ICML)*, 2024. URL <https://arxiv.org/abs/2402.02592>.
- [27] Jiehui Xu et al. Anomaly transformer: Time series anomaly detection with association discrepancy. In *International Conference on Learning Representations (ICLR)*, 2022. URL <https://arxiv.org/abs/2110.02642>.
- [28] Xinyu Zhang et al. Diffusion-ts: Interpretable diffusion for general time series generation. In *International Conference on Learning Representations (ICLR)*, 2024. URL <https://arxiv.org/abs/2403.01742>.
- [29] Haoyi Zhou, Shanghang Zhang, Jieqi Peng, Shuai Zhang, Jianxin Li, Hui Xiong, and Wancai Zhang. Informer: Beyond efficient transformer for long sequence time-series forecasting. In *AAAI Conference on Artificial Intelligence*, 2021.